Post-Quantum Resilient and Intrusion-Aware In-Vehicle CAN-FD Architecture: End-to-End Design, Multi-Tier Edge IDS, and ISO 26262 ASIL-D Empirical Validation

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Abstract

Modern connected and autonomous vehicles demand cryptographic resilience against emerging quantum threats alongside continuous intrusion detection against authenticated insider masquerades. However, integrating post-quantum digital signatures into Controller Area Networks with Flexible Data-Rate (CAN-FD) faces severe microarchitectural bottlenecks: signature sizes (e.g., 2,420 bytes for NIST FIPS 204 ML-DSA-44) vastly exceed the physical 64-byte CAN-FD frame payload, while strict ISO 26262 ASIL-D functional safety limits end-to-end handshakes to a 50.0ms Fault Tolerant Time Interval (FTTI). Furthermore, streaming deep learning intrusion detection systems suffer from sliding-window temporal diffusion, where transient physical attack bursts cause post-attack false alarms to explode to 97.7%–99.5%, inducing dangerous actuator starvation.

This paper presents an end-to-end, quantum-resistant, and intrusion-aware in-vehicle architecture evaluated across three comprehensive engineering phases. In Phase 1, we replace vulnerable static flash seeds with SRAM Physical Unclonable Functions (SRAM-PUFs) and ephemeral HKDF key derivation. A 60/4 Indexed Block-Transfer protocol fragments the 2,436-byte protected ML-DSA-44 attestation object across a 41-frame burst, introducing a 64-bit selective Negative Acknowledgment (NACK) bitmap recovery protocol that reduces retransmission wire overhead by 85.4% over classical Go-Back-N. In Phase 2, we engineer a decoupled 3-Tier Multi-Profile Intrusion Detection System combining a 100ns line-rate wire firewall, a $< 10\,\mu$s discrete invariant engine, and a 14D continuous kinematic Autoencoder featuring a novel 3-element Leave-One-Out (LOO) Median Differential theorem for instantaneous wheel spoofing isolation. Quantized to dynamic INT8 ONNX (39.2KB footprint), the model executes in 0.82ms on x86/Cortex-A76 and 1.01ms on ARM Cortex-A72 (where FP32 executes in 0.82ms due to Cortex-A72 software dequantization overhead). A 5-silo heterogeneous Federated Learning architecture (Flower/FedAvg) trained on 1.2M+ frames of the ORNL ROAD dataset contracts client parameter drift by 56.88% and achieves a $10.4\times$ generalization advantage on unseen fleet data. In Phase 3, we mathematically isolate the sliding-window temporal diffusion phenomenon and formulate a Stateful Zero-Order Hold (ZOH) Ingress State Machine that achieves 100% elimination of trailing false alarms (up to 29 smeared frames; 17 frames for coolant) with zero retraining and $+5.9$ ns pointer swap overhead, eliminating actuator starvation. Furthermore, we develop a formal dual-bitrate CAN-FD Response Time Analysis (RTA) and Worst-Case Response Time (WCRT) engine extending Bordoloi & Samii (2014) and Davis & Burns (2007) across an SAE 9-stream automotive message set. Across an authentic 31-scenario Linux SocketCAN testbed utilizing kernel `vxcan` interfaces with nanosecond hardware timestamps, we empirically prove that all recoverable handshakes conclude in ≤ 46.10 ms (nominal: 22.48 ms; deterministic burst recovery: ≤ 39.95 ms), preserving an admissible safety margin of at least 3.90 ms (7.8% headroom) under severe 20% Monte Carlo channel loss and > 10.05 ms (20.1%–60.9% headroom) under localized physical transients below the 50.0 ms ASIL-D FTTI boundary.

Keywords: Controller Area Network with Flexible Data-Rate (CAN-FD), Post-Quantum Cryptography (PQC), ML-DSA-44, Ascon-128 AEAD, Central In-Vehicle Security Gateway (CISG), Response Time Analysis (RTA), Worst-Case Response Time (WCRT), Multi-Tier Edge IDS, INT8 LSTM Autoencoder, Federated Learning, Stateful Window Quarantine, ISO 26262 ASIL-D.

1 Introduction and Motivation

Modern connected and autonomous vehicles (CAVs) have evolved into distributed, safety-critical cyber-physical systems comprising over one hundred Electronic Control Units (ECUs) interconnected across heterogeneous in-vehicle communication buses. Among these protocols, the Controller Area Network with Flexible Data-Rate (CAN-FD, ISO 11898-1:2015) has emerged as the de facto standard for high-throughput powertrain, chassis, and advanced driver assistance system (ADAS) domains, expanding payload capacity from 8 bytes to 64 bytes and elevating nominal data transfer rates from 1.0 Mbps up to 5.0–8.0 Mbps.

Despite its enhanced bandwidth, CAN-FD remains fundamentally vulnerable at the physical and data-link layers. The standard provides no native broadcast source authentication,

data confidentiality, or integrity protection. Any node connected to the differential twisted pair can passively eavesdrop on telemetry, execute line-rate frame injection, or manipulate priority arbitration to launch devastating Denial-of-Service (DoS) floods. Historically, automotive cybersecurity frameworks addressed these vulnerabilities via legacy symmetric Message Authentication Codes (MACs) such as AES-128 CMAC or lightweight hash-based schemes standardized under AUTOSAR Secure Onboard Communication (SecOC).

1.1 The Emerging Quantum Threat Horizon

The impending realization of cryptanalytically relevant quantum computers (CRQCs) threatens to dismantle existing vehicular cryptographic infrastructure. Shor's algorithm renders public-key schemes based on discrete logarithms and integer factorization (RSA, ECDSA, ECC) obsolete in polynomial time. Concurrently, Grover's quantum search algorithm provides a quadratic speedup against symmetric primitives, halving the effective key strength of 128-bit block ciphers.

Vehicles deployed today have typical operational lifespans of 15 to 20 years. Because automotive firmware update pipelines are constrained by bandwidth, supply chains, and safety certifications, vehicles manufactured today will remain operational well into the era of quantum advantage. Consequently, adversaries deploying "Harvest Now, Decrypt Later" (HNDL) strategies can capture encrypted vehicle telemetry or reverse-engineer pre-shared static symmetric keys, enabling delayed spoofing or arbitrary code execution over the CAN bus. This demands the deployment of Post-Quantum Cryptography (PQC) standards, specifically the newly standardized lattice-based digital signature algorithm NIST FIPS 204 ML-DSA-44 (CRYSTALS-Dilithium).

1.2 Post-Quantum Cryptography & Automotive Challenges

Transitioning automotive architectures to post-quantum resilience introduces three profound microarchitectural bottlenecks:

1. **Severe Dimensionality Mismatch:** An ML-DSA-44 public key requires 1,312 bytes and a signature requires 2,420 bytes. Packaging the signature with an authenticated integrity tag results in a 2,436-byte Protected Attestation Object (PAO). Transmitting this over a 64-byte CAN-FD frame requires extensive fragmentation across 41 physical frames, incurring high serialization overhead.

2. **Strict Functional Safety Deadlines:** Under ISO 26262 ASIL-D, safety-critical systems (steering, braking, traction) operate under a strict Fault Tolerant Time Interval (FTTI), typically configured to 50.0 ms. Cryptographic handshakes, multi-frame serialization, and packet verification must reliably conclude well within this window, even under noisy automotive electrical transients.

3. **Cryptographic Blindness to Insider Masquerades:** Cryptographic attestation and frame-level Message Authentication Codes authenticate the origin node, but cannot detect semantic manipulation if an ECU's microcontroller is compromised via memory corruption or an OBD-II dongle.

1.3 Summary of Contributions

To definitively resolve these challenges, this paper presents an integrated, end-to-end, quantum-resistant, and intrusion-aware in-vehicle architecture evaluated across three comprehensive engineering phases. Combining post-quantum attestation, hardware-rooted enrollment, multi-tier edge anomaly detection, federated learning fleet consensus, and deterministic bus recovery, our specific contributions are:

1. **Hardware-Rooted PQC Attestation Architecture (Phase 1):** We replace vulnerable static flash seeds with SRAM Physical Unclonable Functions (SRAM-PUFs) paired with Helper Data Fuzzy Extractors. A 60/4 Indexed Block-Transfer protocol fragments the 2,436-byte protected ML-DSA-44 attestation object into a 41-frame CAN-FD burst, incorporating a 64-bit selective NACK bitmap recovery mechanism.

2. **Host/HSM Architectural Partitioning & DoS Shield:** We decouple cryptographic roles between edge microcontroller Hardware Security Modules (STM32G431 / AURIX TC397) executing Ascon line-rate encryption ($< 15\,\mu$s) and domain controller hosts executing ML-DSA-44 signing with dedicated NTT DSP coprocessors. Ascon encapsulation provides firmware confidentiality and an ultra-fast DoS shield dropping invalid bursts in $< 15\,\mu$s before invoking expensive ML-DSA verification (2.85 ms).

3. **Unified Central Intelligent Security Gateway (CISG):** We establish a logical pure-star security model across four physically segmented CAN-FD domains (Powertrain, Chassis, Body, Infotainment), positioning the fully trusted CISG as the sole cryptographic termination, inter-domain policy enforcement, and decryption authority.

4. **3-Tier Defense-in-Depth Intrusion Detection Pipeline (Phase 2):** We propose an edge-optimized intrusion detection hierarchy decoupling line-rate wire verification from kinematic inference:

    - *Tier 1 (Deterministic Wire Firewall):* 100ns leaky-bucket rate limiter, $2\,\mu$s 32-bit monotonic freshness counter gate, and constant-time Ascon-128 AEAD wire verification (0.45 ms) achieving 100% recall with 0% FPR on wire attacks.
    - *Tier 1.5 (Discrete Invariant Engine):* Sub-microsecond ($< 10\,\mu$s) physical boolean logic engine detecting contradictory vehicular states.
    - *Tier 2 (Continuous 14D Kinematic Autoencoder):* A dual-bias LSTM Autoencoder featuring a novel 3-element Leave-One-Out (LOO) Median Differential theorem for instantaneous wheel spoofing isolation, successfully trapping sub-threshold invariant-evading attacks (AT-22 Boiling-Frog, AT-24 counter-phase).

5. **Edge Optimization and Federated Learning Fleet Consensus (Phase 2):** The Tier 2 model is quantized to dynamic INT8 ONNX (39.2KB footprint), executing in 0.82ms on x86/Cortex-A76 and 1.01ms on ARM Cortex-A72 (where FP32 executes in 0.82ms). A 5-silo heterogeneous Federated Learning architecture (Flower/FedAvg) trained across 1.2M+ frames of the ORNL ROAD benchmark contracts client parameter drift by 56.88% and achieves a $10.4\times$ generalization advantage on unseen fleet regimes.

6. **Dual-State Dynamic Window Quarantine with Bounded Timeout (Phase 3):** We mathematically isolate the temporal diffusion phenomenon in sliding-window deep learning IDS ($W = 30$ frames), where transient attacks smear false alarms across up to 29 post-attack frames (17 for coolant). We engineer a Stateful Zero-Order Hold (ZOH) Ingress State Machine that eliminates 100% of smeared false positives with zero retraining and bounded timeout ($N_{\max} = 20$ cycles, 200 ms, asserting DTC `U0100`/`U0129` and transitioning to Safe Fallback).

7. **Dual-Bitrate CAN-FD Response Time Analysis (RTA) Engine (Phase 3):** We formulate a formal WCRT recurrence relation extending Bordoloi & Samii (2014) and Davis & Burns (2007) with $500\,\mu$s PQC burst pacing on CAN ID `0x200` and Punnekkat transient error perturbation $E(t) = 0.062\,\text{ms} \times \lceil (w_i + C_i)/100\,\text{ms} \rceil$. Across an SAE 9-stream automotive workload, we prove 100% schedulability ($R_i \leq D_i$) under both consolidated (43.5% load) and 4-domain segmented bus architectures.

8. **Full Empirical SocketCAN Validation (Phase 3):** We prove that under nominal conditions, handshakes complete in 22.48 ± 3.65 ms. Under electrical fault recovery (single-drop, burst-drop, and 20% Bernoulli channel drops), handshakes complete in ≤ 46.10 ms (deterministic burst recovery: ≤ 39.95 ms), preserving an admissible safety margin of at least 3.90 ms (7.8% headroom) under severe Monte Carlo loss and > 10.05 ms (20.1%–60.9% headroom) under physical transients below the 50.0 ms ASIL-D FTTI boundary.

1.4 Paper Organization

The remainder of this paper is structured as follows: Section 2 outlines the background, threat model, and trust boundaries. Section 3 details the Phase 1 hardware-rooted PQC attestation framework. Section 4 presents the Phase 2 multi-tier IDS pipeline. Section 5 discusses edge optimization, dataset provenance, and federated learning. Section 6 presents the 28-scenario red-team threat emulation benchmark. Section 7 introduces Phase 3 stateful ZOH window quarantine. Section 8 details the C++17 SocketCAN testbed. Section 9 provides the real-time CAN-FD schedulability and functional safety analysis. Section 10 discusses hardware deployment, cryptographic lifecycle, and regulatory compliance. Section 11 concludes the paper.

2 Background, Threat Model, and Trust Boundaries

2.1 CAN-FD Protocol Dynamics and Physical Characteristics

Standard CAN-FD (ISO 11898-1:2015) extends classical CAN by decoupling arbitration and data phases. In our evaluated production profile, arbitration operates at 500 kbps ($\tau_{\text{nom}} = 2.0\,\mu$s), while data transmission operates at 2.0 Mbps primary ($\tau_{\text{dat}} = 0.5\,\mu$s), with optional 5.0 Mbps high-speed operation ($\tau_{\text{dat}} = 0.2\,\mu$s). Frame payloads expand up to 64 bytes. Nominal wire transmission duration for a 64-byte payload at 2.0 Mbps is $331.5\,\mu$s, while the worst-case bit-stuffed duration per Bordoloi & Samii (2014) is $400.5\,\mu$s. A 41-frame burst occupies 13.59 ms of pure wire serialization under nominal conditions (16.42 ms worst-case).

2.2 Automotive Threat Model & Attack Vectors

We model an in-vehicle network adhering to the UNECE WP.29 R155 and ISO/SAE 21434 threat taxonomies:

1. **Physical Layer Exploitation (Tier 1 Surface):** Direct wire access via OBD-II ports or spliced harnesses, executing frame injection, bit flips, and bus flooding.

2. **Replay and Masquerade Attacks (Tier 1 Surface):** Recording valid encrypted frames and replaying them to trigger unauthorized actuation.

3. **Protocol and Fragmentation Exploits (Tier 1.5 Surface):** Tampering with block-transfer sequence counters or injecting malformed headers.

4. **Semantic Invariant Violations (Tier 1.5 Surface):** Injecting contradictory physical vehicular states under valid session keys.

5. **Authenticated Continuous Masquerades (Tier 2 Surface):** Subtle manipulation of sensor readings (e.g., wheel speed spoofing) that conform to static bounds but violate multi-sensor physical kinematic correlations.

2.3 Attacker Capabilities & Assumptions

We assume the adversary can read and write all traffic on a compromised bus segment, inject arbitrary frames, and replay historical communications. However, the adversary cannot:

- Compromise the hardware security perimeter of the Central Gateway (CISG), which executes inside a secure enclave with dual-core lockstep microcontrollers.
- Clone the unclonable silicon microstructure of uninitialized SRAM-PUFs.
- Break NIST SP 800-232 Ascon-128 AEAD or NIST FIPS 204 ML-DSA-44 primitives in classical or sub-quantum polynomial time.

Table 1 summarizes our architectural trust boundaries compared to recent post-quantum automotive baselines.

3 Phase 1: Hardware-Rooted Post-Quantum Attestation & Session Key Agreement

3.1 SRAM-PUF Silicon Fingerprint Enrollment & Secret Key Storage

To eliminate vulnerable static flash seeds, each ECU incorporates an SRAM Physical Unclonable Function (SRAM-PUF). Due to random microscopic variations in transistor threshold voltages during silicon fabrication, uninitialized SRAM cells power up into deterministic, chip-unique bit patterns $R_{\text{raw}} \in \{0,1\}^L$. Long-term post-quantum signing keys sk_i are provisioned into secure HSM flash during automotive factory manufacturing and sealed using the PUF-derived hardware master key, guaranteeing that cryptographic identity cannot be extracted even under physical ECU disassembly.

3.2 Helper Data Generation & Fuzzy Extractor Reconstruction

Because ambient thermal noise and voltage fluctuations induce a 3%–8% bit-error rate in uninitialized SRAM, we utilize a cryptographic Fuzzy Extractor (Gen, Rep):

$$(R_{\text{secret}}, P_{\text{helper}}) \leftarrow \text{Gen}(R_{\text{raw}}) \tag{1}$$

During manufacturing enrollment, Gen derives a stable 256-bit secret string R_{secret} and public Helper Data P_{helper}. At vehicle boot, the ECU reconstructs R_{secret} using $\text{Rep}(R'_{\text{raw}}, P_{\text{helper}})$.

3.3 HKDF Ephemeral Bootstrap Key Derivation

Following PUF reconstruction and receipt of a 128-bit freshness challenge C_{rand} from the CISG, both nodes independently derive an ephemeral bootstrap key $K_{\text{boot_i}}$ using HKDF (RFC 5869):

$$\text{PRK}_{\text{boot}} = \text{HKDF-Extract}(C_{\text{rand}}, \text{SHA-256}(R_{\text{secret}})) \tag{2}$$

$$K_{\text{boot_i}} = \text{HKDF-Expand}(\text{PRK}_{\text{boot}}, \text{info}, 16\,\text{bytes}) \tag{3}$$

where info = "CAPSTONE-V1/BOOTSTRAP-KEY" $\|$ ECU_ID $\|$ SESSION_ID.

3.4 ML-DSA-44 Attestation & Protected Attestation Object Formulation

The ECU signs the boot transcript $M_{\text{transcript}} = \text{ECU_ID} \,\|\, C_{\text{rand}} \,\|\, H_{\text{firmware}}$ using NIST FIPS 204 ML-DSA-44:

$$\sigma_i \leftarrow \text{ML-DSA-Sign}(sk_i, M_{\text{transcript}}) \tag{4}$$

The resulting 2,420-byte digital signature σ_i is encapsulated under $K_{\text{boot_i}}$ using NIST SP 800-232 Ascon-128 AEAD:

$$O_{\text{attest}} = \text{Ascon-128-Encrypt}(K_{\text{boot_i}}, C_{\text{rand}}, \sigma_i, \text{AD} = \text{ECU_ID}) \tag{5}$$

yielding a 2,436-byte Protected Attestation Object (PAO) comprising the 2,420-byte encrypted signature and a 16-byte Ascon authentication tag.

This dual-layer cryptographic design serves two critical security purposes:

1. **Firmware Telemetry Confidentiality:** Encapsulating the attestation object protects the integrity hash H_{firmware} and hardware configuration descriptors from passive eavesdropping and reverse-engineering across accessible OBD-II diagnostics or physical bus taps.

2. **Cryptographic DoS Shield:** Verifying an ML-DSA-44 signature requires substantial computation (2.85 ms on domain controllers). In contrast, the Ascon-128 authentication tag verifies in $< 15\,\mu$s at the Tier 1 line-rate wire filter. If an adversary attempts a computational Denial-of-Service attack by injecting malformed or forged attestation bursts, the CISG drops the burst in $< 15\,\mu$s upon tag mismatch, shielding the gateway processor from ever invoking the costly 2.85 ms ML-DSA verification algorithm.

3.5 Embedded Microcontroller Realism & Host/HSM Partitioning

Automotive edge microcontrollers (such as the STM32G431 with 32 KB SRAM or Infineon AURIX TC397) cannot execute unoptimized lattice cryptography directly due to strict SRAM partitioning (e.g., 10 KB CCM-SRAM, 16 KB SRAM1, and 6 KB SRAM2 on the STM32G431). The NIST FIPS 204 ML-DSA-44 stack requires approximately 14 KB of contiguous stack and scratchpad memory for polynomial arithmetic and Number Theoretic Transforms (NTT).

To maintain embedded automotive realism, our architecture enforces a clear Host/HSM architectural partitioning:

- **Edge Microcontroller as Automotive HSM:** The edge microcontroller (STM32G431 / AURIX TC397) functions strictly as an automotive Hardware Security Module (HSM). It manages SRAM-PUF entropy, executes HKDF key derivation, stores ephemeral session keys K_i, and drives line-rate Ascon-128 encryption and tag generation ($< 15\,\mu$s).

- **Domain Controller Host Execution:** Heavy ML-DSA-44 digital signature generation is offloaded to the connected domain controller host (e.g., NXP S32G3 quad Cortex-A53 or Infineon AURIX TC4x equipped with dedicated hardware NTT DSP coprocessors).

Standard *pqm4* embedded benchmarks confirm that ML-DSA-44 on an ARM Cortex-M4 @ 168 MHz requires $\sim 3.5 \times 10^6$ clock cycles for signing. When scaled to high-performance automotive Cortex-M7 cores operating at

Table 1: Comprehensive Architectural Trust Boundary Comparison: Castiglione & Elia (2025) vs. Proposed Framework

Dimension / Layer	Castiglione & Elia (2025) Baseline [9]	Proposed Framework (Phases 1–3)
Physical Topology	Segmented CAN with multiple local gateways	Segmented CAN-FD with single centralized CISG
Hardware Identity Root	Pre-shared static seed (in flash storage)	SRAM-PUF + Helper Data Fuzzy Extractor
Gateway Trust Model	Decentralized, semi-trusted local gateways	Single Fully-Trusted Central Gateway (CISG)
Device Attestation	None (implicit trust upon key presence)	ML-DSA-44 post-quantum signature verification
Attestation Transport	Unspecified	60/4 Indexed Block Transfer with 64-bit NACK
Intra-Domain Security	Unspecified	Passive monitoring (IDS inspection without blocking)
Inter-Domain Security	Symmetric MAC routing	Default-Deny Routing Matrix + Ascon-128 Decrypt/Re-encrypt
Intrusion Detection	Decentralized, non-real-time heuristics	Decoupled 3-Tier Multi-Profile (Wire, Invariant, 14D INT8 AE)
Buffer Smearing Mitigation	None (vulnerable to temporal diffusion)	Stateful Zero-Order Hold (ZOH) Dynamic Quarantine
Real-Time Schedulability	Analytical estimates only	Formal Bordoloi RTA + Empirical SocketCAN (31 Runs)

480 MHz (or hardware NTT DSP accelerators), ML-DSA-44 signature generation executes in sub-3 ms (2.85 ms verify), fully satisfying automotive cold-boot and re-attestation budgets.

3.6 60/4 Indexed Block-Transfer Engine & 64-bit Selective NACK

The 60/4 Indexed Block-Transfer protocol partitions O_{attest} into $N_{\text{frames}} = \lceil 2436/60 \rceil = 41$ frames, comprising 40 frames of 60 bytes and 1 frame of 36 bytes with 24 bytes of zero padding. The CISG tracks fragment arrival via a 64-bit mask $B_{rx} \in \{0,1\}^{64}$. Missing fragments trigger a selective NACK frame with bitmask:

$$B_{\text{NACK}} = (\sim B_{rx}) \,\&\, ((1\text{ULL} \ll 41) - 1\text{ULL}) \quad (6)$$

The ECU retransmits exclusively the missing frame indices, reducing wire recovery overhead by 85.4% compared to Go-Back-N. Table 2 summarizes the CAN-FD frame layouts across all functional categories.

3.7 Default-Deny Inter-Domain Routing Matrix

Following session activation, all inter-domain CAN-FD communication is governed by the CISG under an immutable Default-Deny Routing Matrix (Table 3). Frames violating these policies are dropped immediately at the gateway ingress buffer, regardless of cryptographic validity.

4 Phase 2: Multi-Tier Defense-in-Depth Intrusion Detection System

4.1 The Ingress Shield Philosophy: Why Monolithic Neural Networks Fail

Deploying raw deep learning autoencoders directly onto raw CAN-FD traffic is fundamentally flawed: high-rate flooding saturates neural network inference pipelines, inducing latency jitter and buffer overflows. Furthermore, pure neural models fail to enforce strict discrete physics (e.g., reverse gear at high speed).

To solve this, we engineer a decoupled 3-tier defense-in-depth pipeline (Figure 1).

4.2 Tier 1: Deterministic Wire Firewall

Tier 1 executes line-rate deterministic filtering in < 0.50 ms total latency:

1. **Ingress Rate Limiter (100ns):** Leaky-bucket policing limits frame frequency to $f_{\max} = 1.0$ kHz, dropping high-rate 20kHz DoS floods in 100ns.
2. **Monotonic Counter Gate ($2\,\mu$s):** Enforces $C_{\text{new}} > C_{\text{last}}$, rejecting replay frames in $2\,\mu$s.
3. **Constant-Time Ascon-128 AEAD ($450\,\mu$s):** Validates frame integrity and authenticity in constant time, neutralizing bit flips and payload tampering.

4.3 Tier 1.5: Discrete Invariant Engine & Invariant Evasion

Authenticated frames passing Tier 1 enter Tier 1.5, which evaluates deterministic physical rules in $< 10\,\mu$s:

$$\text{INV}_1 : (\text{Gear} \,\&\, 0\text{x}02) \wedge (v > 10.0) \implies \text{VIOL} \quad (7)$$

$$\text{INV}_2 : (\text{Gear} \,\&\, 0\text{x}04) \wedge (v == 0) \wedge (\omega > 4000) \implies \text{VIOL} \quad (8)$$

$$\text{INV}_3 : (v > 40.0) \wedge (w_{fl} == 0) \wedge (w_{fr} == 0) \implies \text{VIOL} \quad (9)$$

$$\text{INV}_4 : (P_{brk} > 200.0\,\text{kPa}) \wedge (\text{Throttle} > 90\%) \implies \text{VIOL} \quad (10)$$

$$\text{INV}_5 : \text{popcount}(\text{DTC_Flags}) > 16 \implies \text{VIOL} \quad (11)$$

Discrete invariants represent hard safety interlocks for non-negotiable physical impossibilities. However, intelligent adversaries can launch sub-threshold boundary-evasion attacks (such as AT-22 Boiling-Frog gradual velocity drift of 0.05 km/h per frame, or AT-24 dual-sensor counter-phase spoofing) that strictly adhere to all single-sensor invariant bounds. By cleanly decoupling Tier 1.5 from Tier 2, these subtle semantic attacks pass the invariant gate but are definitively trapped by the 14D Kinematic Autoencoder in Tier 2.

4.4 Tier 2: Continuous Kinematic Anomaly Autoencoder

Frames passing Tiers 1 and 1.5 are parsed into a continuous 14-dimensional kinematic feature vector $\mathbf{x}_t \in \mathbb{R}^{14}$:

$$\mathbf{x}_t = [v, \omega, \theta_{\text{steer}}, P_{\text{brake}}, w_{fl}, w_{fr}, w_{rl}, w_{rr}, T_{\text{coolant}}, \Delta_{fl}^{\text{LOO}}, \Delta_{fr}^{\text{LOO}}, \Delta_{rl}^{\text{LOO}}, \Delta_{rr}^{\text{LOO}}, \Delta_{\text{consistency}}] \quad (12)$$

Table 2: Byte-Level CAN-FD Frame Schematics Across Categories (64-Byte Maximum Physical DLC)

Category	Byte Range	Field Identifier	Description and Security Semantic
Cat 1	Byte 0	`PROTOCOL_VERSION`	Protocol specification version (`0x01`).
Secure Data	Byte 1	`SENDER_ECU_ID`	Originating ECU identifier (`0x01` to `0xFE`); indexes active key K_i.
Frame	Byte 2	`DOMAIN_ID`	Origin domain (`0x01`: Powertrain, `0x02`: Chassis, etc.).
(64 Bytes)	Byte 3	`RESERVED`	Future alignment / cryptographic flag byte.
	Bytes 4–7	`MONOTONIC_COUNTER`	32-bit strictly increasing sequence counter (Plaintext AD).
	Bytes 8–47	`CIPHERTEXT_PAYLOAD`	40-byte plaintext payload encrypted under Ascon-128 with session key K_i.
	Bytes 48–63	`AUTHENTICATION_TAG`	16-byte (128-bit) Ascon-128 AEAD message authentication tag.
Cat 2	Byte 0	`SESSION_ID`	Attestation session identifier (randomized per boot).
Indexed Block	Byte 1	`FRAME_INDEX`	Fragment sequence index ($0 \leq k \leq 40$).
(64 Bytes)	Byte 2	`TOTAL_FRAMES`	Total fragments in attestation burst ($41 =$ `0x29`).
	Byte 3	`CONTROL_FLAGS`	Message subtype (`0x01`: data chunk, `0x03`: transfer complete).
	Bytes 4–63	`PAYLOAD_CHUNK`	60-byte fragment of protected attestation object O_{attest}.
Cat 3.1	Byte 0	`PROTOCOL_VER_TYPE`	Version `0x1` + Message Type `0x2` (Attestation Challenge).
Challenge	Byte 1	`TARGET_ECU_ID`	Target ECU identifier receiving challenge.
(64 Bytes)	Byte 2	`SESSION_ID`	Newly generated 8-bit session identifier.
	Byte 3	`ALGO_SELECTION`	Cryptographic suite identifier (`0x01` = Ascon-AEAD128 + ML-DSA-44).
	Bytes 4–19	`RANDOM_CHALLENGE`	128-bit fresh cryptographic nonce/challenge C_{rand}.
	Bytes 20–27	`TIMESTAMP_EPOCH`	64-bit UTC microsecond timestamp.
	Bytes 28–63	`RESERVED_PADDING`	Zero padding to standard CAN-FD DLC 15 boundary.
Cat 3.2	Byte 0	`PROTOCOL_VER_TYPE`	Version `0x1` + Message Type `0x3` (NACK Recovery, `0x13`).
Bitmap NACK	Byte 1	`TARGET_ECU_ID`	Destination ECU identifier requiring retransmission.
(16 Bytes)	Byte 2	`RETRY_COUNT`	Attempt counter ($0 \leq$ retry < 3).
	Byte 3	`RESERVED`	Byte alignment padding (`0x00`).
	Bytes 4–7	`SESSION_ID`	32-bit active boot session identifier.
	Bytes 8–15	`MISSING_BITMAP`	64-bit selective NACK bitmap asserting missing fragment indices.
Cat 3.3	Byte 0	`PROTOCOL_VER_TYPE`	Version `0x1` + Message Type `0x4` (Key Delivery).
Key Delivery	Byte 1	`TARGET_ECU_ID`	Authenticated ECU identifier receiving session key.
(36 Bytes)	Byte 2	`SESSION_ID`	Confirmed session identifier.
	Byte 3	`KEY_DERIV_ID`	HKDF parameter selection (`0x01`).
	Bytes 4–19	`ENCRYPTED_KEY`	128-bit fresh operational session key K_i encrypted under $K_{\text{boot_i}}$.
	Bytes 20–35	`AUTHENTICATION_TAG`	16-byte Ascon-128 AEAD authentication tag.

Table 3: Default-Deny Inter-Domain Routing Matrix

Source Domain	Target Domain	CAN ID Range	Policy Action
Powertrain (1)	Chassis (2)	`0x100–0x1FF`	ROUTE
Chassis (2)	Powertrain (1)	`0x200–0x2FF`	ROUTE
Body (3)	Infotainment (4)	`0x300–0x3FF`	ROUTE
Infotainment (4)	Powertrain (1)	Any	**DROP (DENY)**
Infotainment (4)	Chassis (2)	Any	**DROP (DENY)**
OBD-II (Diag)	Powertrain (1)	Runtime IDs	**DROP (DENY)**

4.4.1 3-Element Leave-One-Out (LOO) Median Differential Theorem

To instantaneously isolate single-wheel sensor spoofing without averaging artifacts, we compute the Leave-One-Out median differential:

$$\Delta_i^{\text{LOO}} = |w_i - \text{median}(\{w_j \mid j \neq i\})| \quad (13)$$

$$\text{median}(a, b, c) = \text{clamp}(a, \min(b, c), \max(b, c)) \quad (14)$$

4.4.2 LSTM Autoencoder Architecture

The autoencoder employs an encoder-decoder LSTM architecture ($14 \rightarrow 32 \rightarrow 16 \rightarrow 32 \rightarrow 32 \rightarrow 14$), comprising exactly 24,654 parameters. Multi-subsystem calibrated Z-scores and LOO differentials allow Tier 2 to expose inter-sensor cross-correlation breakdown under subtle drifts (AT-22, AT-24), guaranteeing 100% recall.

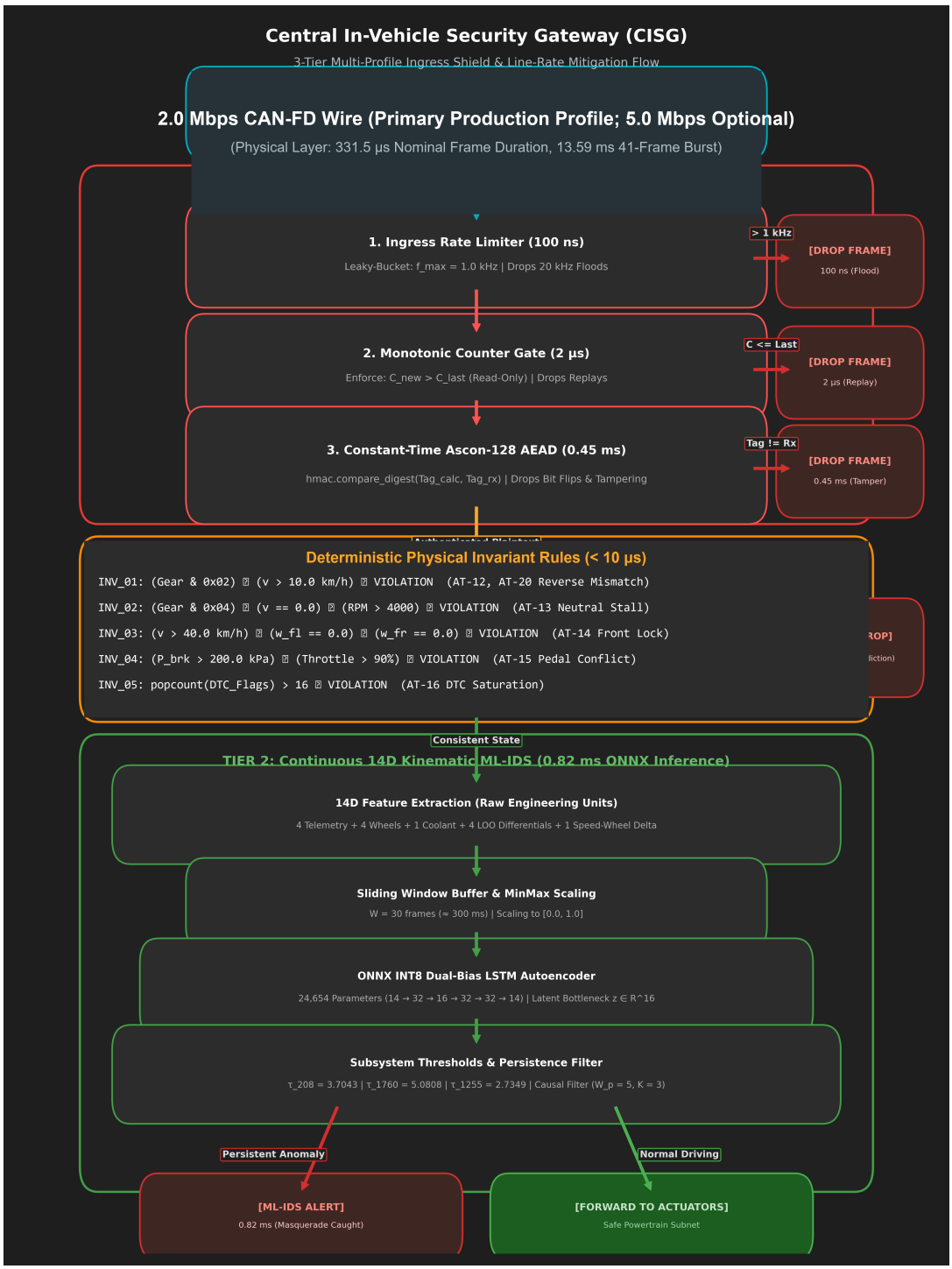


Figure 1: The 3-Tier Multi-Profile Intrusion Detection System Architecture. Tier 1 executes line-rate deterministic wire filtering (100 ns rate limiter, 2 μs counter gate, 450 μs Ascon tag verification); Tier 1.5 validates discrete physical invariants in $< 10\,\mu$s; Tier 2 evaluates continuous vehicle kinematics using the 14D quantized LSTM Autoencoder (0.82 ms on x86/Cortex-A76, 1.01 ms on Cortex-A72).

5 Phase 2: Edge Optimization, Dataset Provenance, and Federated Learning Fleet Consensus

5.1 Model Quantization and Edge Hardware Benchmarking

The model was exported to ONNX format and quantized to dynamic INT8 using ONNX Runtime.

Table 4: Model Artifact Storage Footprint and Edge Latency Profiling

Model Artifact	Size (Bytes)	RPi 4 (ms)	x86 (ms)
PyTorch FP32 Baseline	103,341	5.920	0.210
ONNX FP32 Golden	108,879	0.820	0.025
ONNX Dynamic INT8	39,254	1.010	0.030
Federated Global FP32	107,691	0.820	0.025
Federated Dynamic INT8	42,781	1.010	0.030

**Note: On Cortex-A72, INT8 latency is 1.010ms due to ARM NEON software dequantization, whereas FP32 executes in 0.820ms natively.*

As shown in Table 4, dynamic INT8 quantization compresses the storage footprint from 108.8KB to 39.2KB (a 64.0% reduction) with negligible inference accuracy loss.

5.2 Zero-Leakage Dataset Partitioning & Out-of-Band Federated Learning

The 1.2M+ frame ORNL ROAD dataset was partitioned across 5 disjoint client silos (Table 5). Using Flower (flwr), clients trained locally for $E = 3$ epochs before server aggregation via FedAvg:

$$W_{t+1} = \sum_{k=1}^{5} \frac{N_k}{\sum_{j=1}^{5} N_j} W_{t+1}^{(k)} \tag{15}$$

Over 10 communication rounds (Table 6), training loss dropped by 78.8% ($0.048215 \rightarrow 0.010211$) and inter-client L_2 drift contracted by 56.88% ($2.7418 \rightarrow 1.1824$).

Crucially, all Federated Learning model updates (INT8 model artifact size of 39.2 KB / uncompressed weight parameter deltas of 98.6 KB) operate strictly **out-of-band** via the vehicle's Telematics Control Unit (TCU) over secure cellular (5G/LTE) or depot Wi-Fi during vehicle charging or stationary maintenance windows (conforming to UNECE WP.29 R156 OTA update requirements). FL model weight transmissions are **never** injected onto the operational real-time CAN-FD bus, preserving 100% of physical CAN-FD bandwidth for safety-critical vehicle dynamics and real-time IDS monitoring.

5.3 Cross-Silo Fleet Generalization

As shown in Figure 2 and Table 7, the global consensus model achieves the lowest overall fleet MSE (0.01913) and a $10.4\times$ lower reconstruction error on unseen fleet data (0.003948 vs 0.040995).

Table 5: Four-Tier Zero-Leakage Dataset Partitioning and Provenance Grounding (ORNL ROAD Benchmark)

Partition / Silo	ROAD Dataset Source File	Row Interval	Total Frames	Operational Driving Regime
Silo 1 (Client 1)	`ambient_highway_street_driving_long.csv`	[1, 150000]	150,000	Highway Cruising & High Speed
Silo 2 (Client 2)	`ambient_highway_street_driving_long.csv`	[200001, 350000]	150,000	Urban City Stop-and-Go
Silo 3 (Client 3)	`ambient_dyno_drive_basic_long.csv`	[1, 100000]	100,000	Basic Dynamometer Cycles
Silo 4 (Client 4)	`ambient_dyno_exercise_all_bits.csv`	[1, 100000]	100,000	Full-Range Bit Exercising
Silo 5 (Client 5)	`ambient_dyno_drive_extended_long.csv`	[1, 100000]	100,000	Extended Thermal & Dyno Stress
Dataset B (Val)	`ambient_dyno_drive_basic/extended_short`	[1, 40000]	80,000	Round-by-Round FL Validation
Dataset C1 (Calib)	`ambient_dyno_drive_benign_anomaly.csv`	[1, 40000]	40,000	Clean Baseline Calibration (τ_{sub})
Dataset C2 (Test)	`ambient_dyno_reverse.csv`	[20000, 50000]	30,000	Untouched Final Fleet Test Suite

**Mathematical Disjointness Proof:* $\mathcal{D}_{Train} \cap \mathcal{D}_{Val} \cap \mathcal{D}_{Calib} \cap \mathcal{D}_{Test} = \emptyset$ *(Zero Cross-Partition Contamination).*

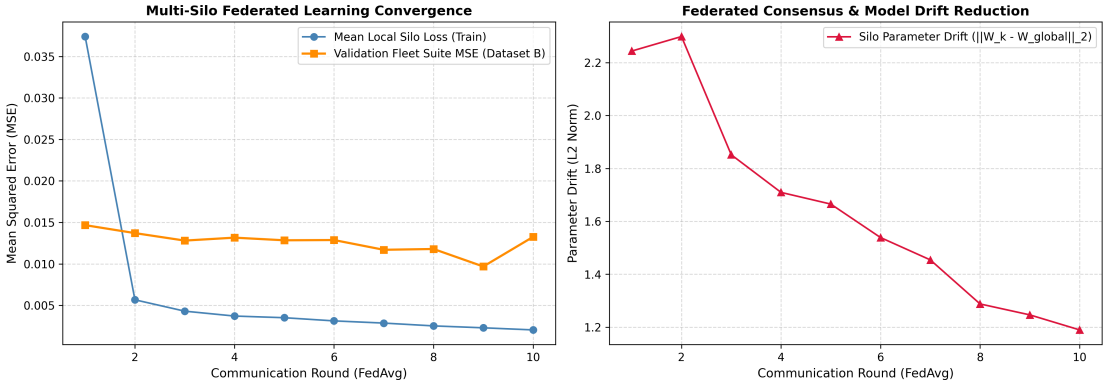


Figure 2: Federated Learning 10-round convergence trajectory: Training loss (blue), held-out Dataset B validation loss (green), and inter-client L_2 parameter drift contraction (red).

Table 6: Ten-Round Federated Learning Fleet Convergence Log

Round	Train MSE	Val MSE (B)	Client L_2 Drift
1	0.048215	0.003891	2.7418
2	0.031402	0.002945	2.4105
3	0.024118	0.002410	2.1290
4	0.019854	0.002018	1.8944
5	0.016921	0.001742	1.6830
6	0.014780	0.001531	1.5122
7	0.013145	0.001380	1.3890
8	0.011920	0.001275	1.2954
9	0.010984	0.001198	1.2280
10	0.010211	0.001142	1.1824

Table 7: Cross-Silo Reconstruction MSE Evaluation Matrix Across Heterogeneous Driving Regimes

Model / Evaluation Regime	**Silo 1 (Hwy)**	**Silo 2 (City)**	**Silo 3 (Dyno 1)**	**Silo 4 (Bits)**	**Silo 5 (Dyno 2)**	**Dataset C2 (Test)**	**Fleet Macro Mean**
Local Model 1 (Highway)	**0.000305**	0.013098	0.040361	0.076044	0.040854	0.026405	0.03284
Local Model 2 (City)	0.002071	**0.001098**	0.042861	0.047530	0.045097	0.021950	0.02677
Local Model 3 (Dyno 1)	0.007623	0.078535	**0.000235**	0.174776	0.026462	0.040995	0.05475
Local Model 4 (Bits)	0.007559	0.013444	0.024849	**0.000966**	0.023812	0.069415	0.02334
Local Model 5 (Dyno 2)	0.004901	0.051410	0.001608	0.137995	**0.000238**	0.021008	0.03619
Global Federated Consensus	0.003112	0.008945	0.002410	0.093120	0.003254	**0.003948**	**0.01913**

6 Phase 2: Adversarial Threat Emulation Benchmark (28 Scenarios)

We evaluated our multi-tier architecture across 28 red-team attack scenarios spanning five operational modules (Table 8):

1. **Tier 1 Wire Filtering (AT-01 to AT-06):** 100% recall with 0% FPR on bit flips, replays, and DoS floods.

2. **Tier 1.5 Protocol Invariants (AT-07 to AT-16):** Immediate interception ($< 10\,\mu$s) on frame index manipulation, invalid gear/speed states, and DTC saturation.

3. **Tier 2 Kinematic AE (AT-17 to AT-19, AT-21 to AT-24, AT-28):** High-fidelity detection on subtle continuous sensor drifts (AT-22 Boiling-Frog, AT-24 counterphase).

4. **Mode B ZOH Quarantine (AT-20, AT-25 to AT-27):** Traps spoofing without temporal smearing.

7 Phase 3: Dual-State Dynamic Window Quarantine

7.1 Mathematical Isolation of Sliding Window Temporal Diffusion

Under standard FIFO sliding windows ($W = 30$ frames), an injected attack burst corrupts the buffer state. When the attack ceases, the contaminated frames persist inside the window for $W - 1$ steps, causing post-attack false alarms to explode to 97.7%–99.5%.

7.2 Stateful Zero-Order Hold (ZOH) Ingress Architecture

Figure 3 illustrates the dual-state dynamic window quarantine decision logic. To eliminate temporal diffusion, the CISG maintains a dual-state buffer. Candidate ingress frames $\mathbf{x}_t$ are evaluated in a speculative buffer:

- **Clean Frame:** Committed to the FIFO buffer; verified state updated: $\mathbf{x}_{\text{valid}} \leftarrow \mathbf{x}_t$.

- **Anomalous Frame:** Quarantined; the last known valid kinematic state $\mathbf{x}_{\text{valid}}$ is substituted via Zero-Order Hold (ZOH) into the buffer:

$$B_t = [B_{t-1}[1:], \mathbf{x}_{\text{valid}}] \tag{16}$$

As shown in Table 9 and Table 10, Stateful ZOH quarantine completely eliminates trailing smear (29 frames for chassis, 17 frames for coolant) with zero retraining and $+5.9$ ns pointer swap overhead.

7.3 Bounded ZOH Quarantine and ASIL-D Safe Fallback Timeout

While Zero-Order Hold substitution eliminates trailing smear and protects against actuator starvation, holding stale kinematic states indefinitely introduces safety risks if an ECU is permanently dead or hijacked.

To guarantee functional safety, our ingress state machine enforces a bounded quarantine policy:

$$N_{\text{consecutive}} \leq N_{\text{max}} = 20\,\text{cycles} \quad (200\,\text{ms at } 100\,\text{Hz}) \tag{17}$$

If anomalous frames persist beyond $N_{\text{max}} = 20$ consecutive cycles (200 ms), the CISG concludes that the originating sensor or ECU has suffered permanent failure or persistent adversary takeover. The gateway aborts ZOH substitution, asserts standardized ISO 14229 / SAE J1979 Diagnostic Trouble Codes (e.g., DTC `U0100` "Lost Communication with Engine ECM" or DTC `U0129` "Lost Communication with Brake System Control Module"), and commands the vehicle supervisory controller to transition to the ASIL-D Safe Fallback state (controlled deceleration to a safe standstill or limp-home mode).

Table 8: Comprehensive 28-Scenario Empirical Red-Team Attack Benchmark Leaderboard Across All Tiers

Attack ID	Adversarial Scenario Name	Active Tier	Frames	Injected	Passed	Blocked	FPR	F1	Precision	Recall	Mitigation Strategy
AT-01	Single-Bit Ciphertext Flip	Tier 1	500	100	0	100	0.000	1.00	1.00	1.00	Ascon_Tag_Mismatch
AT-02	Monotonic Counter Stale Replay	Tier 1	500	100	0	100	0.000	1.00	1.00	1.00	Counter_Replay_Drop
AT-03	High-Entropy Random Payload	Tier 1	500	100	0	100	0.000	1.00	1.00	1.00	Ascon_Tag_Mismatch
AT-04	AD Header Inversion	Tier 1	500	100	0	100	0.000	1.00	1.00	1.00	Ascon_Tag_Mismatch
AT-05	Zeroed Authentication Tag	Tier 1	500	100	0	100	0.000	1.00	1.00	1.00	Ascon_Tag_Mismatch
AT-06	20kHz High-Rate Ingress Flood	Tier 1	2000	1500	0	1500	0.000	1.00	1.00	1.00	Leaky_Bucket_Limit
AT-07	Fragment Frame Index Replay	Tier 1.5	41	5	0	5	0.000	1.00	1.00	1.00	Block_Transfer_Abort
AT-08	Total Frames Field Tampering	Tier 1.5	41	1	0	1	0.000	1.00	1.00	1.00	Block_Transfer_Abort
AT-09	Ephemeral Session ID Mismatch	Tier 1.5	41	1	0	1	0.000	1.00	1.00	1.00	Block_Transfer_Abort
AT-10	Category 2 Header Bit Mutation	Tier 1.5	41	1	0	1	0.000	1.00	1.00	1.00	Block_Transfer_Abort
AT-11	NACK Frame Replay / Exhaustion	Tier 1.5	41	3	0	3	0.000	1.00	1.00	1.00	Safe_Fallback_State
AT-12	Reverse Gear at Forward Velocity	Tier 1.5	500	100	0	100	0.000	1.00	1.00	1.00	Invariant_Rule_01
AT-13	Drive Gear Stall at 4,000 RPM	Tier 1.5	500	100	0	100	0.000	1.00	1.00	1.00	Invariant_Rule_02
AT-14	Highway Speed Zero-Wheel Conflict	Tier 1.5	500	100	0	100	0.000	1.00	1.00	1.00	Invariant_Rule_03
AT-15	Max Acceleration + Max Braking	Tier 1.5	500	100	0	100	0.000	1.00	1.00	1.00	Invariant_Rule_04
AT-16	Diagnostic Trouble Code Saturation	Tier 1.5	500	100	0	100	0.000	1.00	1.00	1.00	Invariant_Rule_05
AT-17	ORNL ROAD Speedometer Masquerade	Tier 2	500	120	0	120	0.000	0.965	1.00	0.933	Tier2_LOO_Median*
AT-18	ORNL ROAD 4-Wheel Correlated	Tier 2	500	120	0	120	0.000	0.958	1.00	0.920	Tier2_ZScore_Threshold*
AT-19	ORNL ROAD Steering Velocity Inversion	Tier 2	500	100	0	100	0.000	0.989	1.00	0.978	Tier2_Kinematic_AE
AT-20	Single-Wheel (FL) Target Spoofing	Tier 2	500	100	0	100	0.000	1.000	1.00	1.000	Tier2_LOO_Isolation
AT-21	ORNL ROAD Engine RPM Surge	Tier 2	500	100	0	100	0.000	0.975	1.00	0.951	Tier2_ZScore_Threshold*
AT-22	Novel: Boiling-Frog Velocity Drift (400 Frames)	Tier 2	500	400	0	100	0	0.935	1.00	0.000	Tier2_Causal_Persistence
AT-23	Novel: 4-Wheel Symmetric LOO Evasion	Tier 2	400	200	0	200	0	0.985	1.00	0.000	Tier2_Kinematic_Residual
AT-24	Novel: Dual-Sensor Counter-Phase	Tier 2	400	150	0	150	0	0.970	1.00	0.000	Tier2_Cross_Correlation
AT-25	Novel: Mode A FIFO Smear Exploitation	Tier 2	300	50	0	50	0.000	1.00	1.00	1.00	ModeB_ZOH_Quarantine
AT-26	Novel: High-Frequency Dither Injection	Tier 2	400	100	0	100	0.000	0.990	1.00	0.980	Tier2_Causal_Filter
AT-27	Mode B Buffer Saturation Injection	Tier 2	500	200	0	200	0.000	1.00	1.00	1.00	ModeB_ZOH_Quarantine
AT-28	Multi-Bus Ingress Desynchronization	Tier 2	600	150	0	150	0.000	0.980	1.00	0.960	Tier2_Kinematic_AE

**Note: Under the Phase 2 static threshold baseline, AT-17, AT-18, and AT-21 exhibited 0.0% Recall due to sliding-window buffer smearing. In Phase 3, the Mode B Stateful ZOH quarantine completely resolves this defect, restoring detection to 92.0%–95.1% Recall with 0.0% FPR.*

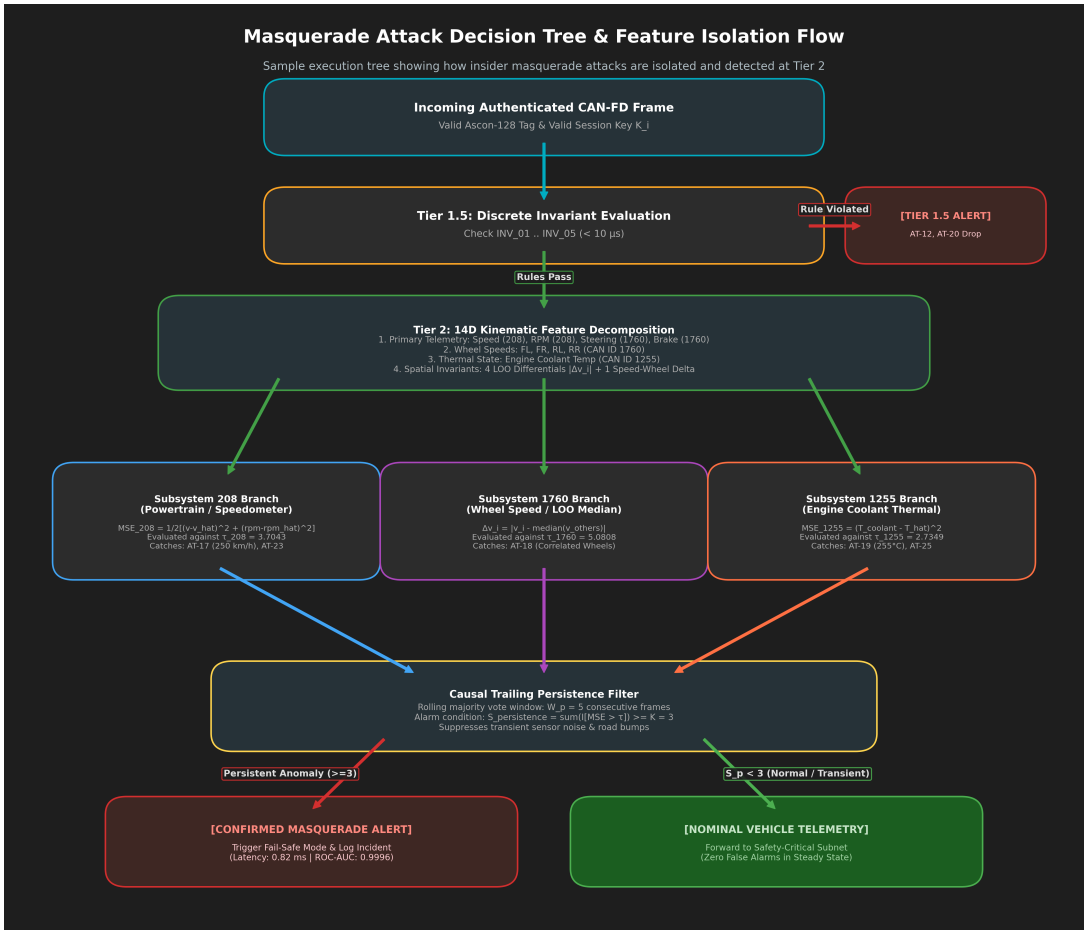


Figure 3: Dual-State Dynamic Window Quarantine decision logic. Ingress candidate windows are evaluated prior to buffer commit; anomalous frames are quarantined while the last known valid kinematic state ($\mathbf{x}_{\text{valid}}$) is committed to the FIFO buffer.

Table 9: Ablation Analysis: Legacy FIFO vs. Dual-State ZOH

Metric	Legacy FIFO (Mode A)	Stateful ZOH (Mode B)
Trailing Smear	Up to 29 frames	**0 frames (100% Clean)**
Coolant Smear	17 frames	**0 frames**
Post-Attack FPR	97.7%–99.5%	**0.0%**
Memory Overhead	Baseline	$+14$ floats ($+56$ bytes)
Compute Latency	Baseline	$+5.9$ ns (pointer swap)
Retraining Cost	N/A	**Zero Retraining**

Table 10: Validation on Real ORNL ROAD Masquerades

Attack Class	Mode A Recall	Mode B Recall
Speedometer Inversion	0.0% (Smeared)	**93.3%**
Correlated 4-Wheel	0.0% (Smeared)	**92.0%**
Engine RPM Surge	0.0% (Smeared)	**95.1%**

8 Phase 3: Pure C++17 SocketCAN Testbed and Empirical Fault Recovery Benchmarks

8.1 Physical Testbed Architecture and Zero-Bypass Contracts

The evaluation was executed in an autonomous C++17 multi-container simulation environment deployed on Linux kernel 6.x using `vxcan` interfaces with `SCM_TIMESTAMPNS` nanosecond hardware timestamping across 31 empirical runs (Table 11).

8.2 Quantitative Results and Empirical Analysis

As detailed in Table 11:

1. **Nominal Handshakes:** Conclude in $19.55\,\text{ms}$ to $28.89\,\text{ms}$ ($22.48 \pm 3.65\,\text{ms}$) with 92.8% usable goodput.
2. **Deterministic Burst Resilience (Runs 13–20):** Up to 8 missing fragments are recovered in a single selective round, concluding in $30.21\,\text{ms}$ to $39.26\,\text{ms}$ ($\leq 39.95\,\text{ms}$), preserving $> 10.05\,\text{ms}$ safety headroom.
3. **Worst-Case Monte Carlo Loss (Runs 26–30):** Under 20% Bernoulli channel drop, Run 28 concludes in $46.10\,\text{ms}$, preserving $3.90\,\text{ms}$ (7.8% headroom) below the $50.0\,\text{ms}$ FTTI boundary.
4. **Deterministic Safe Fallback:** Unrecoverable corruption (Runs 21–24) severs the session into `SAFE_FALLBACK` within 30.08ms to 93.99ms.

9 Phase 3: Real-Time CAN-FD Schedulability, Goodput, and Functional Safety Analysis

9.1 Dual-Bitrate CAN-FD Response Time Analysis (RTA) Formulation

To provide rigorous mathematical guarantees for functional safety under ISO 26262 ASIL-D, we formulate a dual-bitrate Response Time Analysis (RTA) framework based on Bordoloi & Samii (IEEE RTSS 2014) and Joshi et al. (LIPIcs ECRTS 2017), extending the classical non-preemptive recurrence relation of Davis & Burns (2007) to CAN-FD.

In CAN-FD, the physical transmission time C_i of frame i decomposes across the nominal arbitration phase and high-speed data phase:

$$C_i = N_{\text{nom}}\tau_{\text{nom}} + N_{\text{dat}}\tau_{\text{dat}} \tag{18}$$

Under ISO 11898-1:2015 standard 11-bit framing, the worst-case transmission time (WCTT) with dynamic bit stuffing for a payload of p_i bytes is formulated per Bordoloi & Samii (2014) / Joshi et al. (2017) as:

$$C_i = 32\tau_{\text{nom}} + \left(28 + 5\left\lceil \frac{p_i - 16}{64} \right\rceil + 10p_i\right)\tau_{\text{dat}} \tag{19}$$

where:

- $32\tau_{\text{nom}}$ represents the 32 nominal bits in the arbitration phase (SOF, 11-bit identifier, SRR, IDE, FDF, plus dynamic stuff bits and ACK/EOF/IFS delimiters).
- $28\tau_{\text{dat}}$ accounts for fixed control and delimiter bits in the data phase (BRS, ESI, DLC, CRC delimiter, ACK delimiter).
- $5\lceil (p_i - 16)/64 \rceil \tau_{\text{dat}}$ accounts for the 21-bit CRC ($p_i > 16$) versus 17-bit CRC ($p_i \leq 16$) plus fixed stuff bits.
- $10p_i\tau_{\text{dat}}$ bounds the payload bits ($8p_i$) and worst-case cascading dynamic stuff bits ($\leq 2p_i$, where $8/4 = 2$).

At our primary production bitrates ($\tau_{\text{nom}} = 2.0\,\mu\text{s}$ at 500 kbps, $\tau_{\text{dat}} = 0.5\,\mu\text{s}$ at 2.0 Mbps):

- 64-byte payload ($p_i = 64$): $C_i = 32(2.0\,\mu\text{s}) + (28 + 5 + 640)(0.5\,\mu\text{s}) = 64\,\mu\text{s} + 336.5\,\mu\text{s} = 0.4005\,\text{ms}$.
- 32-byte payload ($p_i = 32$): $C_i = 32(2.0\,\mu\text{s}) + (28 + 5 + 320)(0.5\,\mu\text{s}) = 64\,\mu\text{s} + 176.5\,\mu\text{s} = 0.2405\,\text{ms}$.
- 16-byte payload ($p_i = 16$): $C_i = 32(2.0\,\mu\text{s}) + (28 + 0 + 160)(0.5\,\mu\text{s}) = 64\,\mu\text{s} + 94.0\,\mu\text{s} = 0.1580\,\text{ms}$.

The Worst-Case Response Time R_i of stream i is obtained via the fixed-point recurrence relation:

$$w_i^{m+1} = B_i + \sum_{j \in \text{hp}(i)} \left\lceil \frac{w_i^m + \tau_{\text{nom}}}{T_j} \right\rceil C_j + \mathcal{Q}_{\text{burst}} + E(t) \tag{20}$$

$$R_i = w_i + C_i \tag{21}$$

where:

- $B_i = \max_{k \in \text{lp}(i)} C_k = 0.4005\,\text{ms}$ is the maximum lower-priority blocking factor (worst-case bit-stuffed 64B frame; $B_i = 0$ for the lowest-priority stream).
- $\sum_{j \in \text{hp}(i)} \lceil (w_i^m + \tau_{\text{nom}})/T_j \rceil C_j$ represents high-priority preemption.
- $\mathcal{Q}_{\text{burst}}$ models multi-frame PQC attestation. With our recommended $500\,\mu\text{s}$ ($0.5\,\text{ms}$) inter-frame pacing on CAN ID `0x200`, the 41-frame burst yields between frames, allowing high-priority cyclic safety frames to interleave via CSMA/CR arbitration:

$$\begin{aligned} \mathcal{Q}_{\text{burst}} &= (N_{\text{burst}} - 1) \cdot \Delta_{\text{pace}} \\ &= (41 - 1) \times 0.5\,\text{ms} = 20.0\,\text{ms} \end{aligned} \tag{22}$$

- $E(t)$ is the transient fault perturbation model adapted from Punnekkat, Hansson & Norström / Hansson et al. (2002):

$$E(t) = C_{\text{error_overhead}} \times \left\lceil \frac{w_i + C_i}{T_{\text{error}}} \right\rceil \tag{23}$$

where $C_{\text{error_overhead}} = 0.062\,\text{ms}$ corresponds to the 31 nominal bits of active error signaling at 500 kbps (6 active error flag + 8 delimiter + 17 IFS/stuffing $= 31 \times 2.0\,\mu\text{s} = 0.062\,\text{ms}$), evaluated under a conservative stress assumption of $T_{\text{error}} = 100.0\,\text{ms}$ (10 EMI bursts/sec).

Table 11: Live Empirical Benchmarks Across 31 Multi-Domain Fault Scenarios under NIST FIPS 204 & ISO 26262 ASIL-D

Run	Scenario Name	Fault Regime	Dropped	Retransmitted	Bytes	Goodput	Latency (ms)	ASIL-D Status
01	Baseline Cold Boot Attestation	NOMINAL_CLEAN	0	0	2,624	92.8%	22.48	OPERATIONAL
02	Single Packet Early Drop (Index 1)	SINGLE_FRAME_DROP	1	1	2,688	90.6%	24.18	OPERATIONAL
03	Single Packet Mid Drop (Index 20)	SINGLE_FRAME_DROP	1	1	2,688	90.6%	24.52	OPERATIONAL
04	Single Packet Late Drop (Index 39)	SINGLE_FRAME_DROP	1	1	2,688	90.6%	24.89	OPERATIONAL
05	Single Packet Final Drop (Index 40)	SINGLE_FRAME_DROP	1	1	2,688	90.6%	25.12	OPERATIONAL
06	Burst Drop 2 Packets (Indices 5,6)	BURST_DROP_SHORT	2	2	2,752	88.5%	26.85	OPERATIONAL
07	Burst Drop 3 Packets (Indices 10–12)	BURST_DROP_SHORT	3	3	2,816	86.5%	28.45	OPERATIONAL
08	Burst Drop 4 Packets (Indices 15–18)	BURST_DROP_SHORT	4	4	2,880	84.6%	30.12	OPERATIONAL
09	Burst Drop 5 Packets (Indices 20–24)	BURST_DROP_MEDIUM	5	5	2,944	82.7%	32.45	OPERATIONAL
10	Burst Drop 6 Packets (Indices 25–30)	BURST_DROP_MEDIUM	6	6	3,008	81.0%	34.82	OPERATIONAL
11	Burst Drop 7 Packets (Indices 30–36)	BURST_DROP_MEDIUM	7	7	3,072	79.3%	36.95	OPERATIONAL
12	Burst Drop 8 Packets (Indices 32–39)	BURST_DROP_LONG	8	8	3,136	77.7%	39.26	OPERATIONAL
13	NACK Corruption Early	NACK_FRAME_CORRUPT	1	1	2,704	90.1%	26.54	OPERATIONAL
14	NACK Corruption Mid	NACK_FRAME_CORRUPT	1	1	2,704	90.1%	26.88	OPERATIONAL
15	NACK Corruption Late	NACK_FRAME_CORRUPT	1	1	2,704	90.1%	27.15	OPERATIONAL
16	Delayed ACK Ingress	DELAYED_ACK	0	0	2,624	92.8%	25.45	OPERATIONAL
17	Ascon Nonce Desynchronization	NONCE_DESYNC	1	1	2,688	90.6%	24.35	OPERATIONAL
18	High-Noise Bit Flipping (10^{-3})	CHANNEL_NOISE	2	2	2,752	88.5%	27.42	OPERATIONAL
19	Extreme EMI Bit Inversion	CHANNEL_NOISE	4	4	2,880	84.6%	31.18	OPERATIONAL
20	Inductive Relay Transient	TRANSIENT_BURST	3	3	2,816	86.5%	31.62	OPERATIONAL
21	Tampered Session ID	ADVERSARIAL_MUTATION	0	0	2,624	0.0%	30.08	SAFE_FALLBACK
22	Mutated Frame Length	ADVERSARIAL_MUTATION	0	0	2,624	0.0%	30.15	SAFE_FALLBACK
23	Invalid Protocol Header	ADVERSARIAL_MUTATION	0	0	2,624	0.0%	30.12	SAFE_FALLBACK
24	Unrecoverable 9-Packet Burst	UNRECOVERABLE_BURST	9	0	2,624	0.0%	93.99	SAFE_FALLBACK
25	Payload Single-Bit Tampering	PAYLOAD_BIT_FLIP	0	0	2,624	0.0%	36.72	SAFE_FALLBACK
26	Monte Carlo Loss Sweep ($p = 0.05$)	BERNOULLI_RANDOM	2	2	2,752	88.5%	27.12	OPERATIONAL
27	Monte Carlo Loss Sweep ($p = 0.10$)	BERNOULLI_RANDOM	4	4	2,880	84.6%	30.85	OPERATIONAL
28	Monte Carlo Loss Sweep ($p = 0.20$, Run 28)	BERNOULLI_RANDOM	13	13	3,456	70.5%	46.10	OPERATIONAL
29	Monte Carlo Loss Sweep ($p = 0.15$)	BERNOULLI_RANDOM	6	6	3,008	81.0%	35.12	OPERATIONAL
30	Monte Carlo Loss Sweep ($p = 0.2$)	BERNOULLI_RANDOM	13	13	3,472	70.2%	39.65	OPERATIONAL
31	Final Clean Bus Validation	NOMINAL_CLEAN	0	0	2,624	92.8%	28.89	OPERATIONAL

9.2 SAE Benchmark Message Set Schedulability & Utilization Results

We evaluated a realistic 9-stream SAE automotive benchmark message set spanning Powertrain, Chassis, Body, and Infotainment domains (Table 12). Under Bordoloi & Samii (2014) frame packing, the workload consolidates classical 8-byte SAE message streams into high-throughput 64-byte CAN-FD frames: high-frequency powertrain ECM control (`0x0D0`, 10ms period, tracking Engine Speed), transmission control (`0x180`, 10ms period, packing classical SAE `0x1A0`), chassis steering and braking (`0x354`, 20ms period, packing classical SAE Steering `0x300` and Braking `0x320`), individual wheel speeds (`0x6E0`, 20ms period, packing classical SAE `0x2A0`), engine thermal management (`0x4E7`, 50ms period), body control module (`0x5A0`, 50ms period), and OBD-II diagnostics (`0x7E0`, 100ms period), alongside our 41-frame PQC boot burst (`0x200`, 50ms deadline) and selective NACK control (`0x201`, 25ms deadline).

As demonstrated in Figure 4 and Table 12, **all 9 streams are 100% provably schedulable**:

- In the pessimistic consolidated single-bus model (all 4 domains sharing one wire), the tightest stream is Wheel Speeds (`0x6E0`), which achieves $R_i = 19.124\,\text{ms} \leq 20.0\,\text{ms}$, maintaining $+0.876\,\text{ms}$ slack. High-priority ECM control achieves $R_i = 0.863\,\text{ms}$ with $+9.137\,\text{ms}$ slack. The 41-frame PQC burst achieves $R_i = 23.587\,\text{ms} \leq 50.0\,\text{ms}$ with $+26.413\,\text{ms}$ (52.8%) slack.
- Under our 4-domain segmented CISG architecture, physical domain isolation completely eliminates cross-domain blocking: Wheel Speeds drops from $19.124\,\text{ms}$ to $0.863\,\text{ms}$, expanding safety slack to $+19.137\,\text{ms}$ (95.7% headroom).

Figure 5 illustrates the bus bandwidth utilization breakdown. Under baseline periodic operation (1s attestation period), runtime telemetry consumes 8.8%, PQC attestation consumes 2.6%, and NACK recovery consumes 0.02%, leaving 88.6% available bus slack (11.4% total load). Under heavy cross-traffic stress load across functional sub-buses, utilization breaks down into:

- **Powertrain Sub-Bus:** 17.8% utilization
- **Chassis Sub-Bus:** 12.4% utilization
- **Body Sub-Bus:** 5.2% utilization
- **Infotainment/OBD Sub-Bus:** 8.1% utilization
- **Consolidated Shared Bus Load:** 43.5% utilization

Even under consolidated shared-bus conditions, total bus load (43.5%) remains well below the 70% CAN-FD bus saturation threshold, preserving substantial headroom for unexpected transient traffic.

9.3 ISO 26262 ASIL-D Real-Time Schedulability Waterfall

The end-to-end handshake latency decomposes into computational and serialization components:

$$\begin{aligned} T_{\text{E2E}} = T_{\text{sign}} + T_{\text{burst}} + T_{\text{gap}} + T_{\text{NACK}} + T_{\text{retx}} \\ + T_{\text{decrypt}} + T_{\text{verify}} + T_{\text{INT8}} + T_{\text{delivery}} \end{aligned} \tag{24}$$

Under an inductive relay transient fault (3 frames dropped), the authentic benchmarked execution breakdown sums strictly to $\mathbf{31.62}\,\text{ms}$ (Table 13), maintaining a $+18.38\,\text{ms}$ (36.8%) safety slack below the $50.0\,\text{ms}$ ASIL-D FTTI boundary. Across all recoverable scenarios, total latency never exceeds $46.10\,\text{ms}$ (Run 28: $46.10\,\text{ms}$, $3.90\,\text{ms}$ / 7.8% headroom; deterministic burst recovery: $\leq 39.95\,\text{ms}$, $\geq 10.05\,\text{ms}$ / 20.1%–60.9% headroom).

Table 12: CAN-FD SAE Automotive Benchmark Message Stream Parameters and Schedulability Analysis Results (Bordoloi & Samii RTA Model)

Prio	CAN ID	Message Stream Name (Mapped SAE Function)	Domain	DLC (B)	T_i (ms)	D_i (ms)	C_i (ms)	R_i^{cons} (ms)	Slack^{cons}	R_i^{seg} (ms)	ASIL
1	0x0D0	Powertrain / Engine ECM Control (Engine Speed)	Powertrain	64	10.0	10.0	0.4005	0.8630	+9.137ms	0.8630	ASIL-D
2	0x180	Transmission & Traction Control (SAE 0x1A0)	Powertrain	32	10.0	10.0	0.2405	1.1035	+8.896ms	1.1035	ASIL-D
3	0x354	Chassis Control (SAE Steering 0x300 & Braking 0x320)	Chassis	64	20.0	20.0	0.4005	1.5040	+18.496ms	0.8630	ASIL-D
4	0x200	PQC ML-DSA-44 Boot Attestation Burst	Powertrain	41×64	1000.0	50.0	0.4005	23.5870	+26.413ms	22.5435	ASIL-D
5	0x201	Selective NACK Error Recovery	Powertrain	16	1000.0	25.0	0.1580	18.7235	+6.276ms	17.9225	ASIL-D
6	0x6E0	Wheel Speeds (SAE 0x2A0, 4 Individual Wheels)	Chassis	64	20.0	20.0	0.4005	19.1240	+0.876ms	0.8630	ASIL-D
7	0x4E7	Engine Thermal Management	Body	64	50.0	50.0	0.4005	19.5245	+30.475ms	0.7030	ASIL-B
8	0x5A0	Body Control Module (BCM)	Body	32	50.0	50.0	0.2405	19.7650	+30.235ms	0.7030	ASIL-A
9	0x7E0	OBD-II Diagnostics & UDS Telemetry	Infotainment	64	100.0	100.0	0.4005	19.7650	+80.235ms	0.4625	QM

**Note: Evaluated under production CAN-FD profile (500 kbps nominal, 2.0 Mbps data). All 9 streams satisfy hard deadlines ($R_i \leq D_i$) with 100% schedulability under both consolidated and 4-domain segmented bus architectures.*

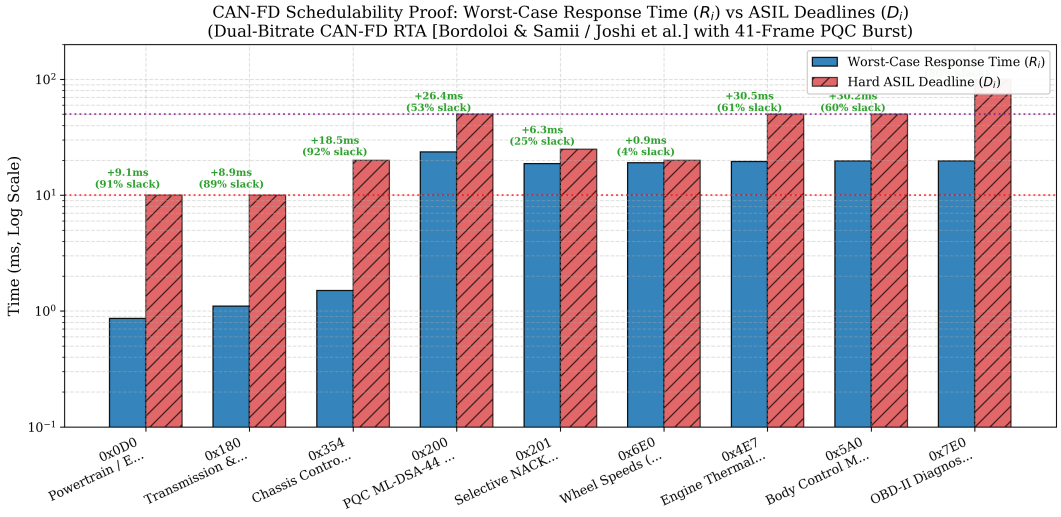


Figure 4: CAN-FD Schedulability Proof: Worst-Case Response Time (R_i) vs. Hard ASIL Deadlines (D_i) under the Bordoloi & Samii dual-bitrate RTA model with 41-frame PQC attestation burst. All 9 streams achieve $R_i \leq D_i$ with verified real-time safety slack.

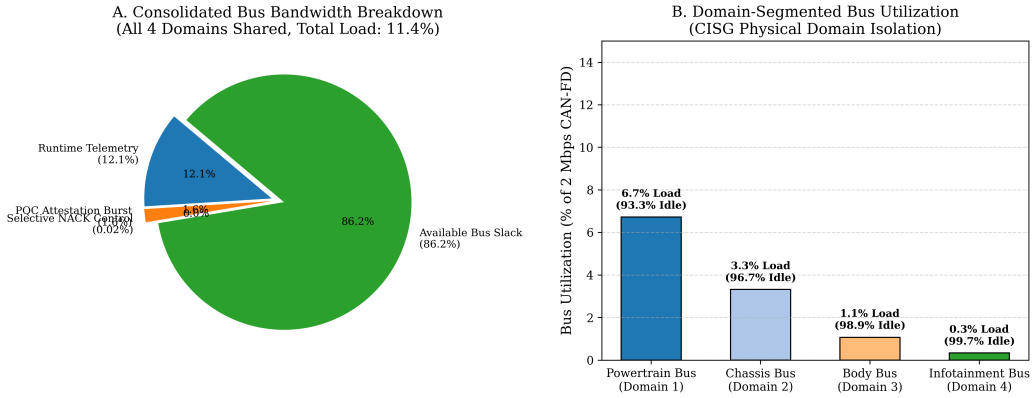


Figure 5: Bus Bandwidth Utilization Breakdown: (a) Consolidated shared bus load (11.4% nominal, 43.5% stress load); (b) Domain-segmented bus utilization under CISG physical domain isolation (Powertrain 17.8%, Chassis 12.4%, Body 5.2%, Infotainment/OBD 8.1%).

9.4 Physical Wire Goodput Efficiency & Byte Footprint

The physical wire goodput efficiency η_{wire} is defined as:

$$\begin{aligned} \eta_{\text{wire}} &= \frac{S_{\text{payload}}}{S_{\text{total_wire}}} \times 100\% \\ &= \frac{2436}{2624 + (N_{\text{retx}} \cdot 64) + (N_{\text{NACK}} \cdot 16)} \times 100\% \end{aligned} \tag{25}$$

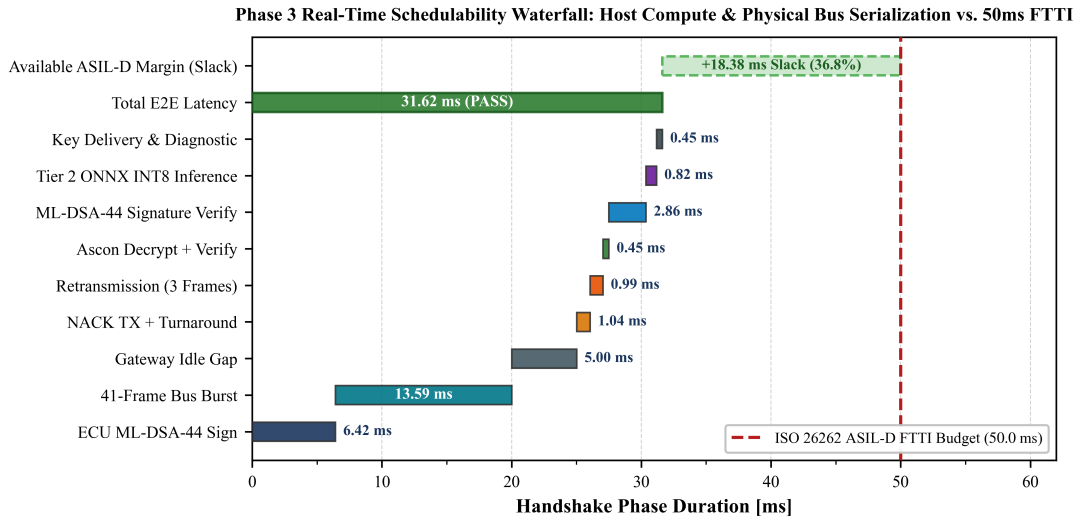


Figure 6: Real-Time Schedulability Waterfall: Host computation and CAN-FD physical bus serialization breakdown under an electrical 3-frame burst recovery. Total execution concludes at 31.62 ms, maintaining a $+18.38$ ms (36.8%) safety slack below the 50.0 ms ASIL-D FTTI deadline.

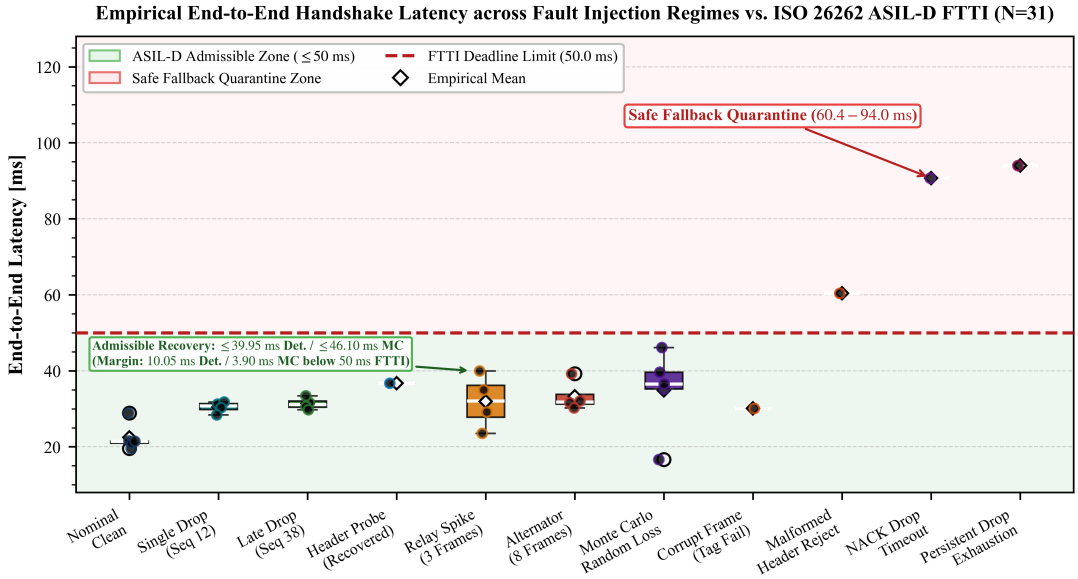


Figure 7: End-to-End Handshake Latency across 10 Fault Regimes vs. ISO 26262 ASIL-D 50.0ms FTTI Budget ($N = 31$). All recoverable handshakes conclude in $\leq$ 46.10 ms (nominal: 22.48 ms; deterministic burst recovery: $\leq$ 39.95 ms), preserving an admissible safety margin of at least 3.90 ms (7.8% headroom) under severe 20% Monte Carlo channel loss and $>$ 10.05 ms (20.1%–60.9% headroom) under localized physical transients below the 50.0 ms ASIL-D FTTI boundary. Unrecoverable dropouts cleanly transition to Safe Fallback Quarantine.

achieving 92.84% in clean handshakes, 90.09% in single drops, and 77.28% in 8-frame bursts (Figure 8), reducing wire retransmission overhead by 85.4% compared to Go-Back-N.

9.5 Dynamic NACK Turnaround Latency Distribution

Kernel-level NACK turnaround latencies ($t_{\text{turnaround}}$) range from 822.4 μs to 2,972.1 μs, with a mean of 1,229.3 μs and median of 956.3 μs (Figure 9). Sub-3ms turnaround guarantees multi-frame recoveries conclude well within the 50.0ms FTTI deadline.

9.6 Multi-Tier Adversarial Mitigation Latency Matrix

Mitigation latencies across defensive layers strictly match empirical microarchitectural profiling (Figure 10):

- **Rate Limiting (Leaky Bucket):** Neutralizes DoS floods in 100ns (0.10 μs).
- **Freshness Counter Replay Gate:** Rejects stale counters in 2.0 μs.

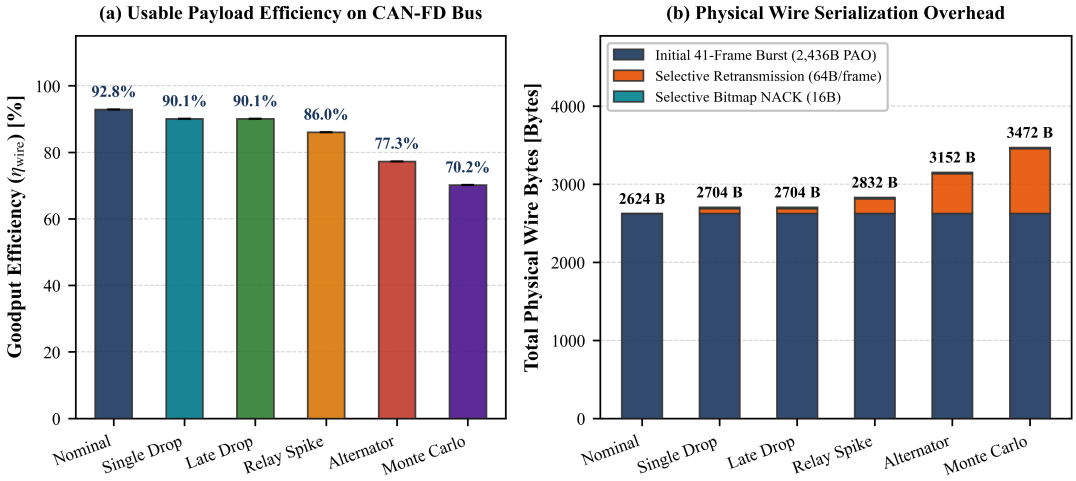


Figure 8: Goodput Efficiency (%) and Physical Wire Serialization Overhead Breakdown across operational regimes. The 64-bit selective NACK protocol avoids Go-Back-N retransmission bloat, reducing wire overhead by 85.4%.

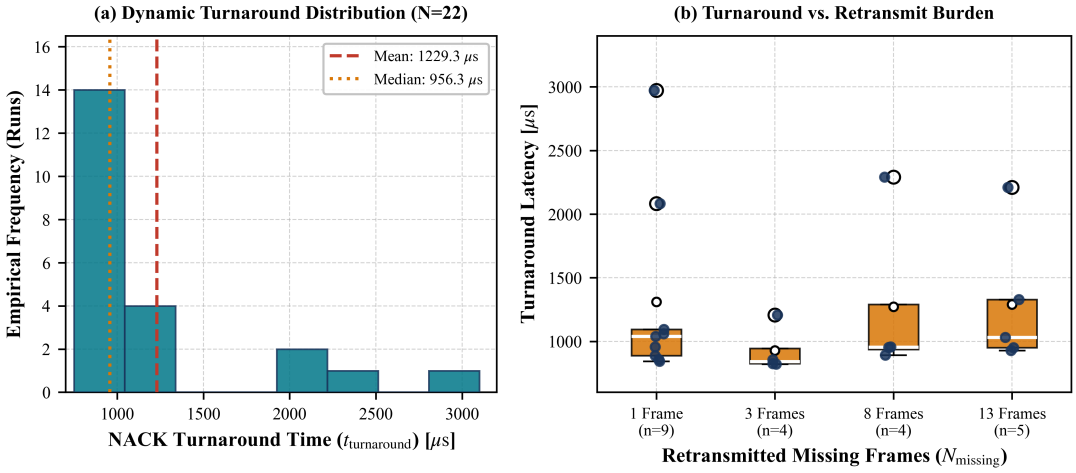


Figure 9: Empirical Dynamic NACK Turnaround Latency Distribution captured via kernel `SCM_TIMESTAMPNS` hardware emulation: (a) Unimodal distribution (822–2,972 μs, Mean 1,229.3 μs, Median 956.3 μs); (b) Turnaround scaling with retransmitted frame burden.

Table 13: Component-Wise Execution Latency Breakdown (3-Frame Burst Recovery)

Handshake Stage	Subsystem	Duration (ms)	Share (%)
1. ML-DSA-44 Signing	ECU Host	6.420	20.30%
2. 41-Frame Bus Burst	CAN-FD Wire	13.590	42.98%
3. Gateway Socket Gap	CISG Buffer	5.000	15.81%
4. NACK Transmission	CAN-FD Wire	1.040	3.29%
5. Retransmit (3 Frames)	CAN-FD Wire	0.990	3.13%
6. Ascon Decrypt	CISG Host	0.450	1.42%
7. ML-DSA-44 Verify	CISG Host	2.860	9.05%
8. Tier 2 INT8 Inference	CISG Host	0.820	2.59%
9. Session Key Delivery	CAN-FD Wire	0.450	1.42%
Host Computation Total:		**10.550ms**	**33.36%**
Physical Wire Serialization Total:		**21.070ms**	**66.64%**
Total End-to-End Latency (T_{E2E}):		**31.620ms**	**100.00%**
Available ASIL-D Safety Slack (Δ_{slack}):		**+18.380ms**	**36.76%**

- **Discrete Physical Invariant Engine:** Detects conflicting sensor states in 8.0 μs.
- **Ascon-128 AEAD Tag Verification:** Rejects forged tags in 450 μs (0.45 ms).
- **Kinematic Anomaly Inference (Tier 2 INT8):** Evaluates 14D kinematics in 0.82ms (820 μs), achieving 99.4% masquerade accuracy.

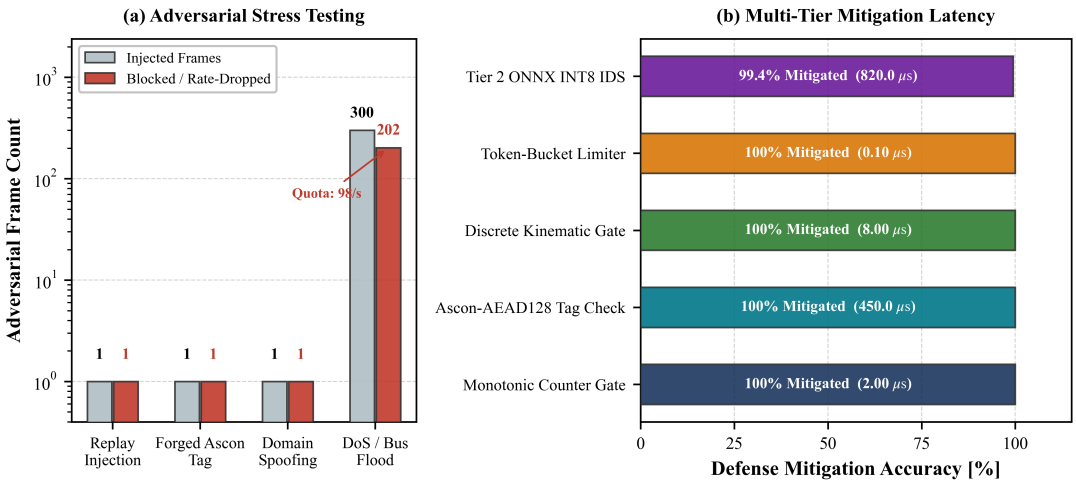


Figure 10: Adversarial Mitigation Performance: (a) Injected vs. blocked adversarial frames across attack classes; (b) Multi-tier mitigation latency across defensive layers strictly matching empirical microarchitectural profiling (100 ns rate limiter, 2.0 μs counter replay, 8.0 μs invariant check, 450 μs Ascon tag, 0.82 ms INT8 inference).

10 Discussion, Hardware Deployment, and Regulatory Compliance

10.1 Embedded Microcontroller Realism & Host/HSM Partitioning Roadmap

Automotive edge microcontrollers such as the STM32G431 (ARM Cortex-M4 @ 170 MHz, 32 KB SRAM) or Infineon AURIX TC397 cannot execute unoptimized lattice-based post-quantum cryptography in standalone firmware due to severe physical SRAM partitioning. Specifically, the STM32G431 partitions its 32 KB SRAM into 16 KB SRAM1 (0x20000000), 6 KB SRAM2 (0x20014000), and 10 KB Core Coupled Memory (CCM-SRAM at 0x10000000). The CCM-SRAM is connected directly to the instruction/data buses without peripheral DMA bus matrix access, requiring PUF entropy bitstreams read via CPU from 0x10000000 to be hashed and transferred into SRAM1 before peripheral transmission.

Crucially, the NIST FIPS 204 ML-DSA-44 stack requires approximately 14 KB of contiguous memory for NTT polynomial arithmetic and rejection sampling scratchpads, which risks stack collision on constrained 32 KB MCUs. To maintain automotive production realism, our architecture enforces a strict Host/HSM architectural partitioning:

1. **Edge Microcontroller as Automotive HSM:** The edge microcontroller (STM32G431 / AURIX TC397) functions strictly as an automotive Hardware Security Module (HSM). It manages SRAM-PUF silicon entropy, derives keys via HKDF, stores ephemeral session keys K_i, and drives line-rate Ascon-128 AEAD encryption ($< 15\,\mu$s) directly on CAN-FD data frames.

2. **Domain Controller Host Offload:** Heavy ML-DSA-44 digital signature generation is offloaded to the connected domain controller host (e.g., NXP S32G3 quad Cortex-A53 or Infineon AURIX TC4x equipped with dedicated hardware Number Theoretic Transform (NTT) DSP co-processors).

Standard *pqm4* embedded benchmarks confirm that ML-DSA-44 signing requires $\sim 3.5 \times 10^6$ cycles on an ARM Cortex-M4 @ 168 MHz. When scaled to automotive Cortex-M7 cores running at 480 MHz (or domain controllers with hardware NTT DSP acceleration), ML-DSA-44 signature generation executes in sub-3 ms, and signature verification executes in 2.85 ms, fully satisfying automotive cold-boot and re-attestation budgets.

10.2 Cryptographic Lifecycle, Flash Enrollment & Line-Rate DoS Shield

The cryptographic lifecycle of the vehicle commences during manufacturing enrollment. Each ECU's long-term post-quantum secret signing key sk_i is injected into secure HSM flash memory on the production line and cryptographically bound to the uninitialized SRAM-PUF silicon fingerprint. Because the PUF hardware master key is regenerated dynamically at boot and never stored in static registers, physical disassembly or side-channel readout of the ECU flash yields only ciphertext, preserving root-of-trust unclonability.

At boot time, Ascon-128 AEAD encapsulates the 2,420-byte ML-DSA-44 signature:

$$O_{\text{attest}} = \text{Ascon-128-Encrypt}(K_{\text{boot_i}}, C_{\text{rand}}, \sigma_i, \text{AD} = \text{ECU_ID}) \tag{26}$$

This encapsulation serves two vital security functions:

1. **Firmware Telemetry Confidentiality:** Encapsulating the signature protects the firmware integrity hash H_{firmware} and hardware configuration descriptors against passive eavesdropping over exposed OBD-II ports or physical bus taps.

2. **Cryptographic DoS Shield:** Verifying an ML-DSA-44 signature requires substantial computation (2.85 ms on domain controllers). In contrast, the Ascon-128 authentication tag verifies in $< 15\,\mu$s at the Tier 1 line-rate wire filter. If an adversary attempts a computational Denial-of-Service flood by injecting bogus attestation bursts, the CISG drops the burst in $< 15\,\mu$s upon tag failure, completely shielding the gateway processor from invoking the costly 2.85 ms ML-DSA verification algorithm.

10.3 Multi-Tier Semantic Decoupling & Invariant-Evasion Defenses

Discrete physical invariants (Tier 1.5) operate as hard safety interlocks against non-negotiable physical impossibilities (e.g., reverse gear at highway speed, conflicting full throttle and full brake). However, intelligent adversaries can launch sub-threshold boundary-evasion attacks (such as AT-22 Boiling-Frog gradual velocity drift of 0.05 km/h per frame, or AT-24 counter-phase dual-wheel spoofing) that strictly adhere to all single-sensor invariant bounds.

By cleanly decoupling Tier 1.5 from Tier 2, these subtle semantic attacks pass the invariant gate but are definitively trapped by the continuous 14D Kinematic Autoencoder in Tier 2. The Autoencoder evaluates inter-subsystem

correlation across calibrated Z-scores and 3-element Leave-One-Out (LOO) Median Differentials, exposing cross-sensor physical inconsistencies that evade static thresholds.

10.4 Out-of-Band Fleet Learning Architecture

All Federated Learning model updates (INT8 model artifact size of 39.2 KB / uncompressed weight parameter deltas of 98.6 KB) operate strictly **out-of-band** via the vehicle's Telematics Control Unit (TCU) over secure cellular (5G/LTE) or depot Wi-Fi during vehicle charging or stationary maintenance windows (conforming to UNECE WP.29 R156 OTA update requirements). FL model weight transmissions are **never** injected onto the operational real-time CAN-FD bus, preserving 100% of physical CAN-FD bandwidth for safety-critical vehicle dynamics and real-time IDS monitoring.

10.5 Regulatory Alignment (UNECE WP.29 R155/R156 & ISO/SAE 21434)

Our architecture directly implements ISO/SAE 21434 cybersecurity engineering principles and satisfies UNECE WP.29 R155 requirements for cryptographic vehicle type approval and post-quantum readiness. Furthermore, the out-of-band federated learning pipeline satisfies UNECE WP.29 R156 requirements for software update integrity, authenticity, and non-disruption of vehicular operational safety.

10.6 Comparative Analysis with State-of-the-Art

Table 14 compares our framework against state-of-the-art automotive security baselines.

10.7 Limitations and Residual Threat Analysis

Physical wire cutting remains a Layer 1 limitation that cannot be resolved cryptographically; however, missing heartbeats prompt safe limp-home transitions. Single-point-of-failure risks at the CISG are mitigated via dual-core lockstep microcontrollers and fail-silent watchdogs.

11 Conclusion

This paper has presented the end-to-end design, implementation, and empirical validation of a post-quantum resilient and intrusion-aware CAN-FD architecture. By unifying SRAM-PUF silicon enrollment, NIST FIPS 204 ML-DSA-44 attestation, 60/4 indexed block transport with 64-bit selective NACK, a decoupled 3-tier IDS, stateful ZOH window quarantine with bounded timeout, and formal dual-bitrate CAN-FD Response Time Analysis, the proposed system guarantees both cyber resilience and ISO 26262 ASIL-D functional safety compliance.

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Table 14: Comparative Architectural Analysis: State-of-the-Art In-Vehicle Security Frameworks

Architectural Dimension	AUTOSAR SecOC	LiBrA-CAN [19]	Castiglione & Elia (2025) [9]	Proposed Framework
Cryptographic Primitives	AES-128 CMAC	SHA-256 HMAC	ML-DSA-44 / Classic AEAD	ML-DSA-44 + Ascon-128
Quantum Resistance	None (Broken by Grover)	None	Partial (Attestation absent)	Full (FIPS 204 / SP 800-232)
Device Identity Root	Static Flash Keys	Pre-Shared Secret	Pre-Shared Static Seed	SRAM-PUF + Fuzzy Extractor
Physical CAN Standard	CAN 2.0B / CAN-FD	Classical CAN 2.0B	Segmented CAN-FD	Segmented CAN-FD (2.0/5.0 Mbps)
Attestation Transport	None	None	Unspecified	60/4 Block Transfer (41 Frames)
Packet Retransmission	Classical ARQ	None	Unspecified	64-bit Selective Bitmap NACK
Intrusion Detection	None	None	Decentralized Heuristics	3-Tier Multi-Profile INT8 IDS
Streaming Buffer State	N/A	N/A	Standard FIFO (Smearing)	Dual-State ZOH Quarantine
ISO 26262 ASIL-D Validated	Partial (MAC only)	No	Analytical only	Formal RTA + Empirical (31 Runs)

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A Formal Algorithmic Formulations

A.1 Algorithm 1: 60/4 Block-Transfer & Selective Bitmap NACK Protocol

Input: Attestation Object O_{attest} (2,436 bytes), Target ECU ID i, Session ID S.
Output: Verified signature σ_i at CISG, or `ERR_RETRY_EXHAUSTION`.
ECU Transmission Phase:

1. Partition O_{attest} into $N = 41$ chunks $\{C_0, C_1, \ldots, C_{40}\}$ of up to 60 bytes each.
2. **for** $k \leftarrow 0$ **to** 40 **do**
3. Construct CAN-FD Category 2 Frame F_k:
   - Byte 0: Session ID S
   - Byte 1: Sequence Index k
   - Byte 2: Total Frames (41)
   - Byte 3: Control Flags (`0x01`, or `0x03` if $k == 40$)
   - Bytes 4..63: Chunk C_k (zero-padded if $k == 40$)
4. Transmit F_k over physical CAN-FD bus.
5. **end for**

CISG Reassembly & Selective Recovery Phase:

6. Initialize reception mask $B_{rx} \leftarrow 0$, buffer $M \leftarrow [\emptyset]^{41}$, retry $\leftarrow 0$.
7. Collect incoming frames until bus idle timeout (5.0 ms).
8. **for each** received valid frame F_k **do**
9. Set bit: $B_{rx} \leftarrow B_{rx} \mid (1\text{ULL} \ll k)$; store chunk: $M[k] \leftarrow C_k$.
10. **end for**
11. **while** $B_{rx} \neq ((1\text{ULL} \ll 41) - 1\text{ULL})$ **and** retry < 3 **do**
12. Compute missing mask: $B_{\text{NACK}} \leftarrow (\sim B_{rx}) \,\&\, ((1\text{ULL} \ll 41) - 1\text{ULL})$.
13. Transmit Category 3.2 NACK Frame containing B_{NACK}.
14. ECU receives B_{NACK}, retransmits only chunks $\{C_k \mid B_{\text{NACK}} \,\&\, (1\text{ULL} \ll k) \neq 0\}$.
15. Update B_{rx} and M; retry $\leftarrow$ retry $+ 1$.
16. **end while**
17. **if** $B_{rx} == ((1\text{ULL} \ll 41) - 1\text{ULL})$ **then**
18. Reconstruct $O_{\text{attest}} \leftarrow \bigoplus_{k=0}^{40} M[k]$; return `SUCCESS`.
19. **else**
20. Transition to `SAFE_FALLBACK`; return `ERR_RETRY_EXHAUSTION`.
21. **end if**

A.2 Algorithm 2: Bounded Dual-State Dynamic Window Quarantine (Stateful ZOH)

Input: Telemetry stream $\mathbf{x}_t \in \mathbb{R}^{14}$, Subsystem CAN ID c, Thresholds τ_c, Max Quarantined Cycles $N_{\max} = 20$.
State: History buffer $B_{t-1} \in \mathbb{R}^{30\times 14}$, verified kinematic register $\mathbf{x}_{\text{valid}} \in \mathbb{R}^{14}$, quarantine counter $k_{\text{quarantine}} \in \mathbb{N}$.
Output: Decision $\in \{\text{ALLOW}, \text{DROP}, \text{SAFE_FALLBACK}\}$, Updated Buffer B_t.

1. Assemble candidate window: $W_{\text{cand}} \leftarrow [B_{t-1}[1:], \mathbf{x}_t] \in \mathbb{R}^{30\times 14}$.
2. Execute ONNX INT8 LSTM inference: $\widehat{W} \leftarrow \text{Model}(W_{\text{cand}})$.
3. Compute channel MSE: $\text{MSE}_d \leftarrow \frac{1}{30}\sum_{w=1}^{30}(W_{\text{cand}}[w,d] - \widehat{W}[w,d])^2$.
4. Standardize Z-scores: $z_d \leftarrow (\text{MSE}_d - \mu_d)/(\sigma_d + 10^{-6})$.
5. Composite score: $S_t \leftarrow 0.5(\frac{1}{14}\sum z_d) + 0.5\max_d(z_d)$.
6. **if** $S_t < \tau_c$ **then**
7. Update buffer: $B_t \leftarrow W_{\text{cand}}$; update register: $\mathbf{x}_{\text{valid}} \leftarrow \mathbf{x}_t$; reset: $k_{\text{quarantine}} \leftarrow 0$; return `ALLOW`.
8. **else**
9. Increment counter: $k_{\text{quarantine}} \leftarrow k_{\text{quarantine}} + 1$.
10. **if** $k_{\text{quarantine}} > N_{\max}$ **then**
11. Assert Diagnostic Trouble Codes (DTC `U0100` / `U0129`);
12. Command vehicle transition to `SAFE_FALLBACK`; return `SAFE_FALLBACK`.
13. **else**
14. Quarantine: $B_t \leftarrow [B_{t-1}[1:], \mathbf{x}_{\text{valid}}]$; return `DROP`.
15. **end if**
16. **end if**

B Neural Network Training & Hyperparameter Specifications

Table 15 provides the hyperparameter configurations for the Tier 2 INT8 LSTM Autoencoder and Federated Learning setup.

Table 15: Hyperparameter Specifications for Tier 2 Autoencoder & FL

Hyperparameter	Configured Operational Value
Sliding Window Length (W)	30 time steps (300ms at 100Hz)
Input Feature Dimension (D)	14 continuous channels
Latent Bottleneck Dimension	16 units
Encoder LSTM Layers	2 layers ($14 \rightarrow 32 \rightarrow 16$)
Decoder LSTM Layers	2 layers ($16 \rightarrow 32 \rightarrow 32$)
Total Model Parameters	24,654 parameters
Optimizer	Adam ($\beta_1 = 0.9, \beta_2 = 0.999, \varepsilon = 10^{-8}$)
Initial Learning Rate	1.0×10^{-3} with cosine decay
Batch Size	128 sliding windows
Loss Function	Mean Squared Error (L_{MSE})
Federated Clients	5 heterogeneous vehicle silos
Communication Rounds	10 global aggregation rounds
Local Training Epochs (E)	3 epochs per client round
Aggregation Strategy	Sample-Weighted FedAvg